REPUBLIC OF THE PHILIPPINES
PROVINCE OF NUEVA ECIJA
CITY GOVERNMENT OF CABANATUAN

A Web-Based AI-Assisted Ordinance Information System for Cabanatuan City

Comprehensive Multi-Role User & Administrative Manual
End-User Citizen • Barangay Administrator • LGU Super Administrator

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📌 NOTE: HOW TO USE THIS OPERATIONAL MANUAL

This user manual is formatted on a per-user-role basis. To perform tasks specific to your official capacity, navigate directly to your designated chapter:

- Public Citizens: Chapter 4 (Workflows 4.1 to 4.7)
- Barangay Captains & Secretaries: Chapter 5 (Workflows 5.1 to 5.7)
- City Hall & LGU Super Administrators: Chapter 6 (Workflows 6.1 to 6.10)

Each section contains high-resolution genuine application screenshots with numbered callout circles corresponding exactly to the step-by-step instructions.

Chapter 1: Executive Summary & System Overview

The Cabanatuan City Ordinance Portal is an integrated web-based civic technology platform engineered to modernise, centralise, and automate municipal and barangay ordinance governance across the City of Cabanatuan, Nueva Ecija. Operating across eighty-nine (89) administrative barangays, local government units historically faced fragmented physical records, cumbersome paper drafting procedures, manual OCR encoding bottlenecks, and limited public visibility into enacted local laws.

To decisively overcome these governance challenges, this project introduces a unified, multi-tiered legislative ecosystem combining zero-barrier civic transparency for citizens, localized jurisdictional autonomy for barangay councils, and comprehensive executive oversight for City Hall.

1.1 Core Innovation Matrix

System Innovation	Technical Implementation & Impact
Multimodal AI Document Extractor	Powered by Google Gemini multimodal vision processing. Government clerks upload scanned PDF copies or smartphone photos of signed physical ordinances. The engine performs high-precision OCR, strips bureaucratic legal preambles, and populates resolution numbers, series years, categories, coverage, and tiered fine schedules with human-in-the-loop review in under 5 seconds.
Bilingual Gemini AI Civic Assistant (RAG)	A conversational Retrieval-Augmented Generation assistant embedded on the public portal. Citizens inquire in English, Tagalog, or Taglish (e.g., 'Ano ang parusa sa illegal dumping sa Camp Tinio?'). The pipeline injects verified local database legislation into the LLM context window to provide grounded, citation-rich answers while strictly refusing to give out-of-scope legal advice.
Multi-Tenant Barangay Isolation	Complete data partitioning and RBAC enforcement. Barangay administrators can only draft, view, and modify legislative records and community incident reports within their assigned jurisdiction, while City Hall LGU Super Admins retain consolidated city-wide oversight.
End-to-End Incident Traceability	A closed-loop community reporting mechanism. Residents submit violation complaints with automatic location landmarking and draft auto-save persistence. The system routes the report to the local Barangay Tanod desk, provides an audit reference tracking code to the citizen, and updates resolution statuses in real time.

Chapter 2: System Architecture & Role Permissions

The application is engineered on Next.js 16 (App Router & Turbopack), React 19, Tailwind CSS v4, Prisma ORM 7 with SQLite, and the Google GenAI SDK. Authentication is enforced through cryptographically signed JSON Web Tokens (jose) stored in Secure HTTP-only cookies, eliminating client-side token interception risks.

User permissions adhere strictly to the principle of least privilege through three dedicated roles:

2.1 Role-Based Access Control (RBAC) Matrix

System Module / Capability	Citizen / Public	Barangay Admin	LGU Super Admin
Public Landing Page & 3D Archive	Full Access (No Login)	Full Access	Full Access
Ordinance Explorer & PDF Viewer	Full Access (No Login)	Full Access	Full Access
AI Civic Assistant (Chatbot RAG)	Full Access (No Login)	Full Access	Full Access
Community Incident Report Submission	Full Access (Draft Persistence)	Full Access	Full Access
Barangay Governance Dashboard	No Access (Denied)	Full Access (Scoped to Barangay)	Supervisory View
Drafting Local Barangay Ordinances	No Access (Denied)	Full Access (Save Draft / Submit)	Full Access (City Enactments)
AI Document OCR Extraction Tool	No Access (Denied)	Local Extraction & Upload	Full City Hall Extraction
Review & Approve/Reject Submissions	No Access (Denied)	No Access (Barangay is Submitter)	Full Legislative Authority
User & Personnel Account Management	No Access (Denied)	No Access (Barangay Level)	Full Admin User Provisioning
Barangay Directory & Coverage	View Public Directory	View Local Barangay	Full 89 Barangay Management
Incident Report Resolution Triage	No Access (Resident View)	Triage Local Reports (Tanod Desk)	City-Wide Oversight & Auditing

Civic News Publishing & Broadcast	View Public Feed	Propose Local Advisories	Publish, Edit, Pin City Bulletins
Official Legislative Report Generator	No Access	Barangay Local Summary PDF	Full Municipal PDF Generator
Executive Governance Analytics	No Access	Barangay Resolution Rates	Comprehensive Recharts Analytics

PRO-TIP: SECURITY ARCHITECTURE COMPLIANCE

The system complies with Section II of the BSIT Capstone Defense Guide. Passwords are never stored in plaintext and are hashed using bcrypt with 10 salt rounds. Multi-tenant database queries enforce WHERE barangayId = session.barangayId on all mutation endpoints, preventing unauthorized cross-jurisdiction access.

Chapter 3: System Access & Multi-Role Authentication

Access to the Cabanatuan City Ordinance Portal is structured into two tiers: zero-barrier public exploration (requiring zero registration or login) and secure administrative access for city and barangay officials.

3.1 Public Portal Landing & Civic Navigation

Step 1: Accessing the Portal — Open any modern web browser (Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari) and navigate to the official portal domain (or http://localhost:3000 for local administrative installations).

Step 2: Public Portal Navigation — Inspect the civic navigation bar (Badge 2) containing direct links to Ordinances, Community Reports, AI Assistant, Civic News, and Administrative Sign-in.

Step 3: Keyword Quick Search — Locate the prominent Hero Search input (Badge 3). Type in any keyword such as 'Traffic', 'Basura', or 'Curfew' to execute immediate full-text search across all municipal legislation.

Step 4: Category Discovery — Click any of the quick discovery category chips (Badge 4) to instantly filter ordinances by municipal interest areas.

Step 5: Interactive 3D Archive Exploration — Interact with the Three.js 3D Ordinance Archive Canvas (Badge 5) using mouse drag or touch gestures to rotate the cylindrical archive.

Step 6: Toggling Accessibility Themes — Click the Theme Toggle button (Badge 6) in the upper right header to switch between Light and Dark mode.

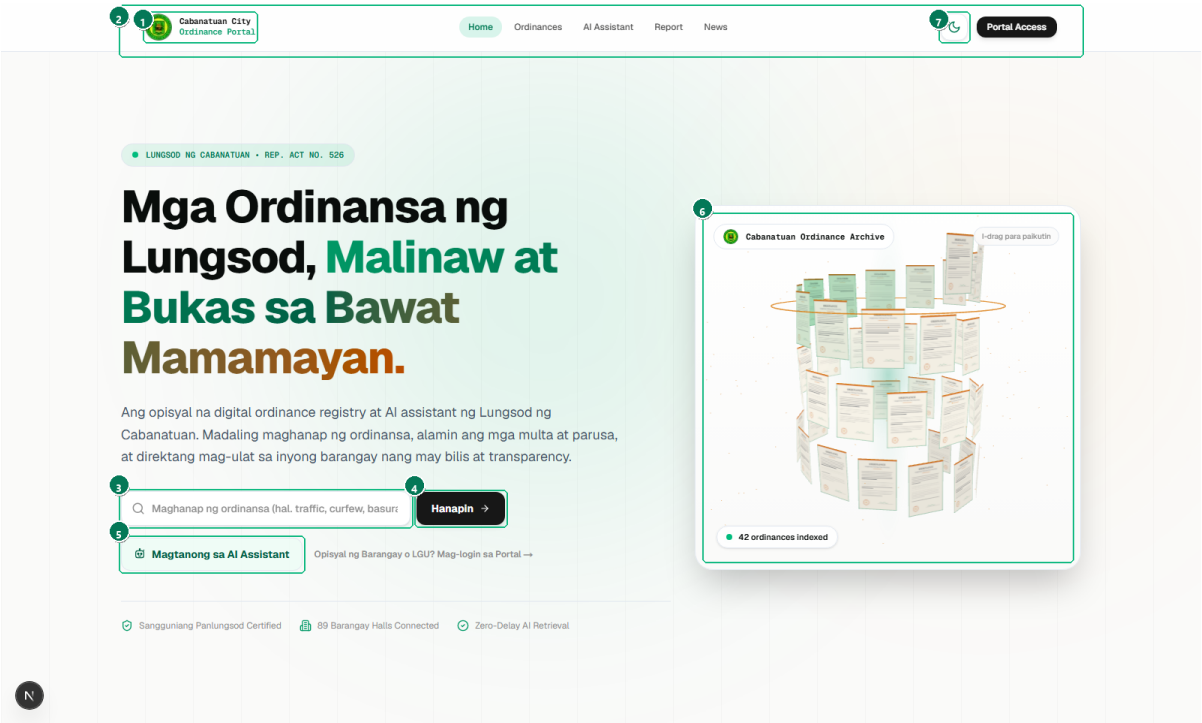


Figure 3.1: *The public homepage provides zero-barrier access to city ordinances, quick search, interactive 3D archive, and civic links.*

#	Interface Element / Control	Operational Purpose & Usage

1	**Cabanatuan City Seal & Portal Branding**	Official seal of the City Government of Cabanatuan, linking back to the root landing interface.
2	**Main Navigation Bar**	Top persistent navigation bar hosting direct links to Ordinances, Report, AI Assistant, News, and Login.
3	**Hero Ordinance Search Input**	Primary full-text query bar supporting real-time keywords, resolution numbers, and Tagalog terms.
4	**Search Submit Button**	Search submission trigger initiating instant client-side filtering across approved city legislation.
5	**Ask AI Assistant Action Button**	1-click shortcut launching the bilingual Cabanatuan Ordinance AI citizen conversational assistant.
6	**Interactive 3D Ordinance Archive Canvas**	Interactive WebGL Three.js 3D cylindrical visualization displaying active municipal legislation.
7	**Portal Theme Toggle (Dark/Light)**	Accessibility theme toggle allowing instant switching between high-contrast Dark Mode and crisp Light Mode.

3.2 Multi-Role Authentication Procedure

Step 1: Navigating to Authentication — Click the 'Mag-log In' button on the navigation bar, or navigate directly to /login.

Step 2: Entering Official Email — In the Email field (Badge 2), enter your verified administrative email address (e.g., admin@cabanatuan.gov.ph for LGU Super Admin or kapitan@cabanatuan.gov.ph for Barangay Admin).

Step 3: Entering Password — In the Password field (Badge 3), input your secure password. The interface masks characters for shoulder-surfing protection.

Step 4: Submitting Credentials — Click the 'Pumasok sa Portal' button (Badge 4). Upon verification, the system creates an HTTP-only JWT session cookie and redirects you automatically to your designated dashboard.

Step 5: Demo Preset Quick-Fill — For defense demonstrations or training, click any of the preset role chips (Badge 5) to auto-fill verified demonstration credentials.

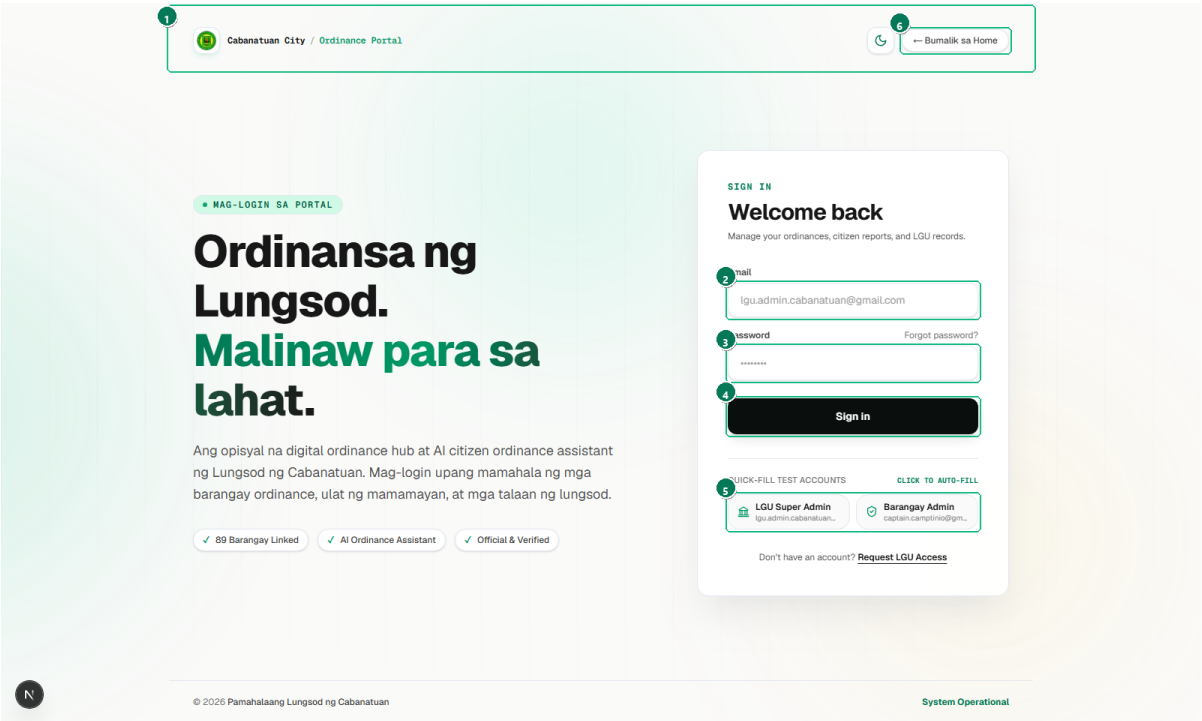


Figure 3.2: *Secure sign-in interface supporting Role-Based Access Control (RBAC) for LGU Super Administrators, Barangay Officials, and Citizens.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Official Portal Authentication Header**	Official civic authentication branding with municipal security badges and credential encryption notice.
2	**Official Email Address Input Field**	Email entry field accepting registered administrative credentials (e.g., admin@cabanatuan.gov.ph).
3	**Password Input Field**	Masked password field authenticated via salted bcrypt hashing and HTTP-only session tokens.
4	**Sign In / Submit Button**	Primary authentication submission button triggering server-side role evaluation and session dispatch.
5	**Quick Role Preset Access Chips**	Rapid test-credential selector enabling instant sign-in for demonstration and testing purposes.
6	**Return to Public Portal Homepage Link**	Public redirect anchor allowing users to exit the authentication boundary and return to public browsing.

3.3 High-Contrast Dark Mode & Accessibility

Step 1: Engaging Dark Theme — Click the Theme Toggle icon (Badge 2). The application dynamically swaps CSS design tokens across backgrounds, borders, and typography.

Step 2: Reviewing High Contrast — Inspect the high-contrast dark header and search bar (Badges 1 and 3) designed with muted emerald accents for night viewing.

Step 3: Visualizing 3D Elements in Dark Mode — The WebGL canvas (Badge 4) automatically adapts its lighting shaders and particle background to dark aesthetics.



Figure 3.3: *Zero-flicker dark aesthetic compliant with WCAG accessibility standards, designed for comfortable nighttime civic reading.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Dark High-Contrast Civic Header**	Dark-optimized civic header with emerald branding maintaining AAA contrast standards.
2	**Active Dark Mode Theme Toggle**	Active state theme switch reflecting dark mode engagement across the entire application shell.
3	**Dark Glassmorphic Search Bar**	Glassmorphic high-contrast search input designed for glare-free nighttime legibility.
4	**Interactive 3D Cylinder Archive (Dark Scheme)**	Three.js canvas dynamically re-rendered with ambient emerald glow and dark cylindrical facets.

PRO-TIP: OFFICIAL DEMONSTRATION CREDENTIALS REFERENCE

For defense evaluation, the system is seeded with three primary verified accounts:

- LGU Super Admin: admin@cabanatuan.gov.ph | Password: admin123 (Redirects to /admin/lgu)
- Barangay Admin (Camp Tinio): kapitan@cabanatuan.gov.ph | Password: kapitan123 (Redirects to /admin/barangay)
- Resident / Citizen: juan@cabanatuan.gov.ph | Password: citizen123 (Public exploration & report filing)

Chapter 4: Citizen & Public End-User Guide

A cornerstone of the Cabanatuan City Ordinance Portal is zero-barrier public civic transparency. Cabanatuan residents, students, business owners, and legal researchers can browse enacted municipal regulations, examine certified PDF copies, consult the bilingual Gemini AI Civic Assistant, and submit incident reports without mandatory account creation.

4.1 Public Ordinance Explorer & Multi-Filtering

Step 1: Navigating to Explorer — Click 'Mga Ordinansa' in the navigation bar or visit /ordinances.

Step 2: Instant Keyword Filtering — Enter terms into the Full-Text Search box (Badge 1). The interface filters records instantly with zero page reloads.

Step 3: Selecting Policy Categories — Click any category button (Badge 2) such as 'Kapayapaan at Kaayusan', 'Kalusugan', or 'Kapaligiran' to narrow results.

Step 4: Adjusting Year Slider — Drag the Year Range Slider (Badge 3) to isolate legislation enacted within specific legislative calendar terms.

Step 5: Filtering by Jurisdiction — Use the Jurisdiction Scope buttons (Badge 4) to toggle between City-Wide Ordinances and Barangay Regulations.

Step 6: Browsing Results — Click any ordinance card in the responsive grid (Badge 5) to inspect its full legal text.

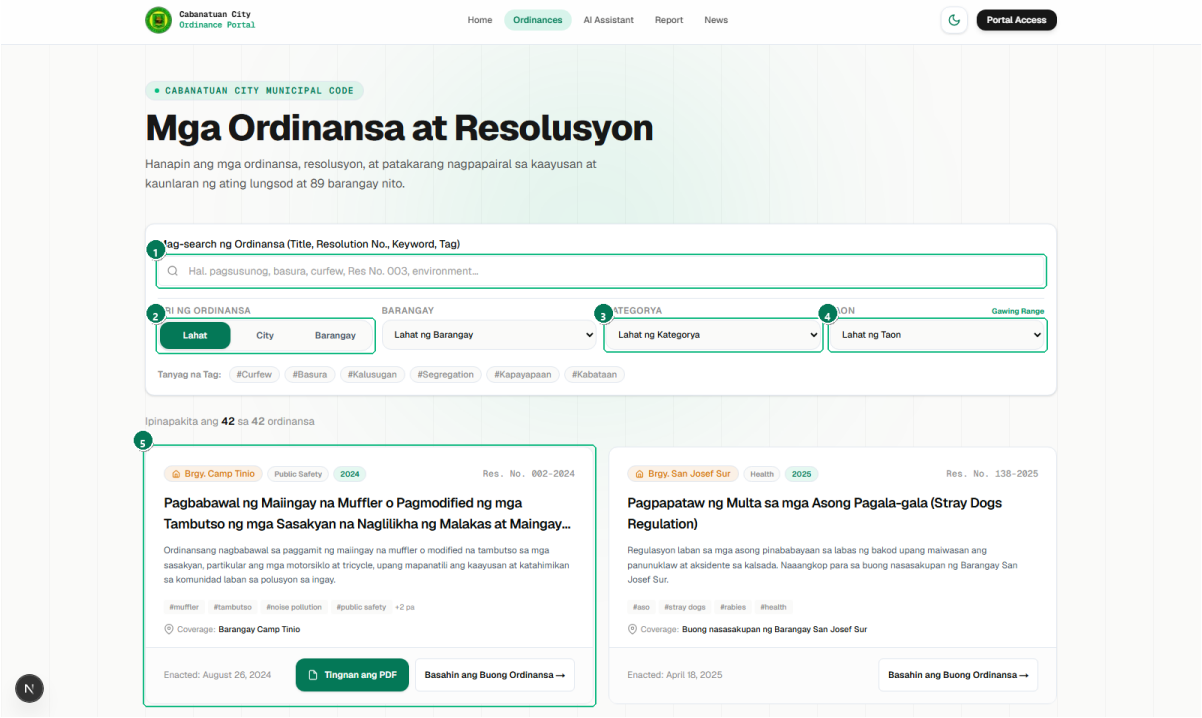


Figure 4.1: *Comprehensive search and multi-filtering system enabling citizens to find city and barangay ordinances by keyword, year, and category.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Full-Text Keyword Search Input**	Real-time debounce search input filtering ordinances across title, resolution number, and content.

2	**Jurisdiction Scope Filter (Lahat, City, Barangay)**	Jurisdiction filter segmented buttons toggling between City-Wide and Barangay-level ordinances.
3	**Policy Category Filter Dropdown**	Policy category selector dropdown filtering ordinances by municipal administrative domain.
4	**Legislative Year Filter Dropdown**	Legislative year selector dropdown isolating ordinances enacted in specific legislative terms.
5	**Ordinance Card Grid with Status & Category Tags**	Responsive card grid displaying approved ordinances with title, category, year, and read-more actions.

4.2 Split-Screen Legal Text & Sanctions View

Step 1: Reviewing Heading & Resolution Number — Verify the official ordinance title and unique legislative resolution number at the top of the view (Badge 1).

Step 2: Inspecting Legislative Metadata — Check the enacting date, year series, and authorizing body (Badge 2).

Step 3: Reading Legal Provisions — Read through the structured legal articles and prohibited acts on the split-screen left pane (Badge 3).

Step 4: Inspecting Sanctions & Penalties — Examine the dedicated Sanctions Card on the right sidebar (Badge 4) for tiered monetary fines and community service provisions.

Step 5: Accessing Certified PDF Document — Click the 'Buksan ang Opisyal na PDF' button (Badge 5) to open the verified scanned original document.

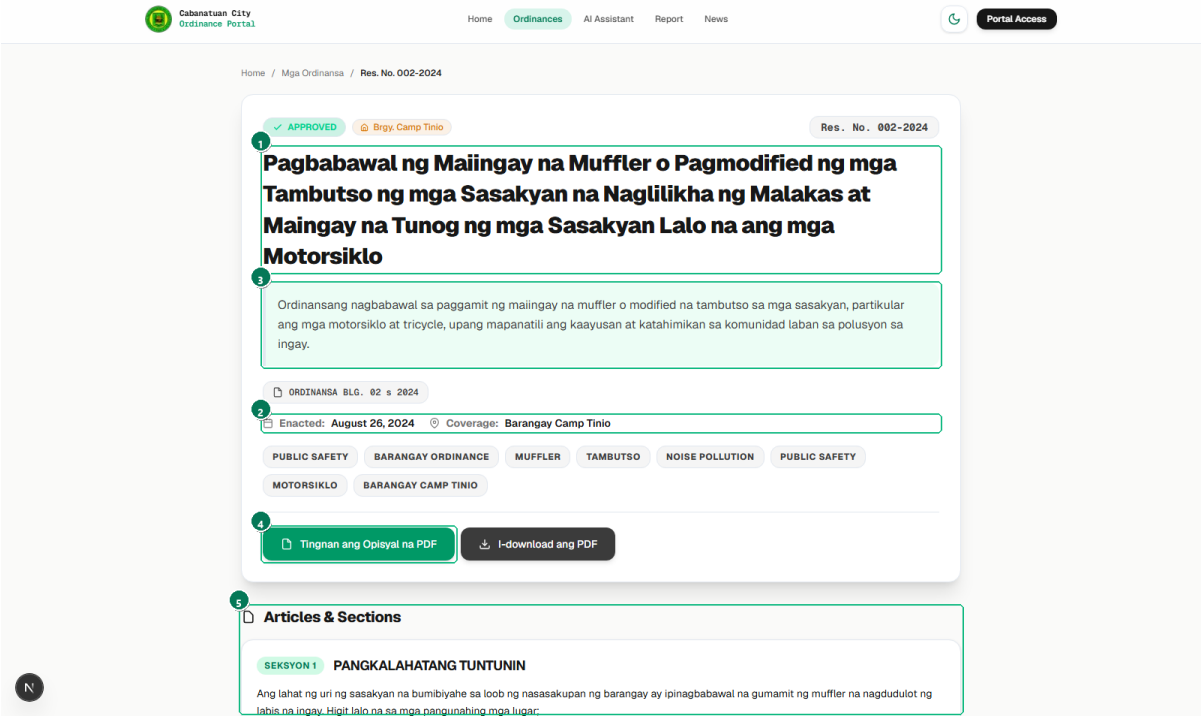


Figure 4.2: *Structured presentation of ordinance text, breaking complex legislation into readable articles, penalties, and enforcement mechanisms.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Official Ordinance Title & Resolution Number**	Official ordinance heading and enacted resolution identifier formatted for civic clarity.
2	**Legislative Metadata Bar (Date Enacted & Territorial Coverage)**	Comprehensive enactment metadata including date passed, series year, and presiding legislative body.
3	**Executive Legal Summary & Background Rationale Box**	Executive legal summary box providing plain-language overview of the legislative mandate.
4	**Open Official Scanned PDF Viewer Action Button**	High-priority action button launching the interactive scanned PDF document modal dialog.
5	**Structured Legal Articles & Sections Container**	Structured legal provisions broken down into numbered articles, sections, and prohibited acts.

4.3 Official Scanned PDF Viewer Modal

Step 1: Inspecting Scanned Resolution Copy — The embedded viewer (Badge 2) loads the scanned physical document complete with official dry seals and physical signatures.

Step 2: Using Interactive Zoom Controls — Use the Zoom In (+) and Zoom Out (-) buttons (Badge 3) to inspect fine print, signatures, and legal annotations.

Step 3: Downloading the PDF File — Click the 'I-download ang Dokumento' button (Badge 4) to save a certified copy to your computer or mobile device.

Step 4: Closing the Modal Dialog — Click the Close button (Badge 5) or press the Escape key to return to the ordinance overview.

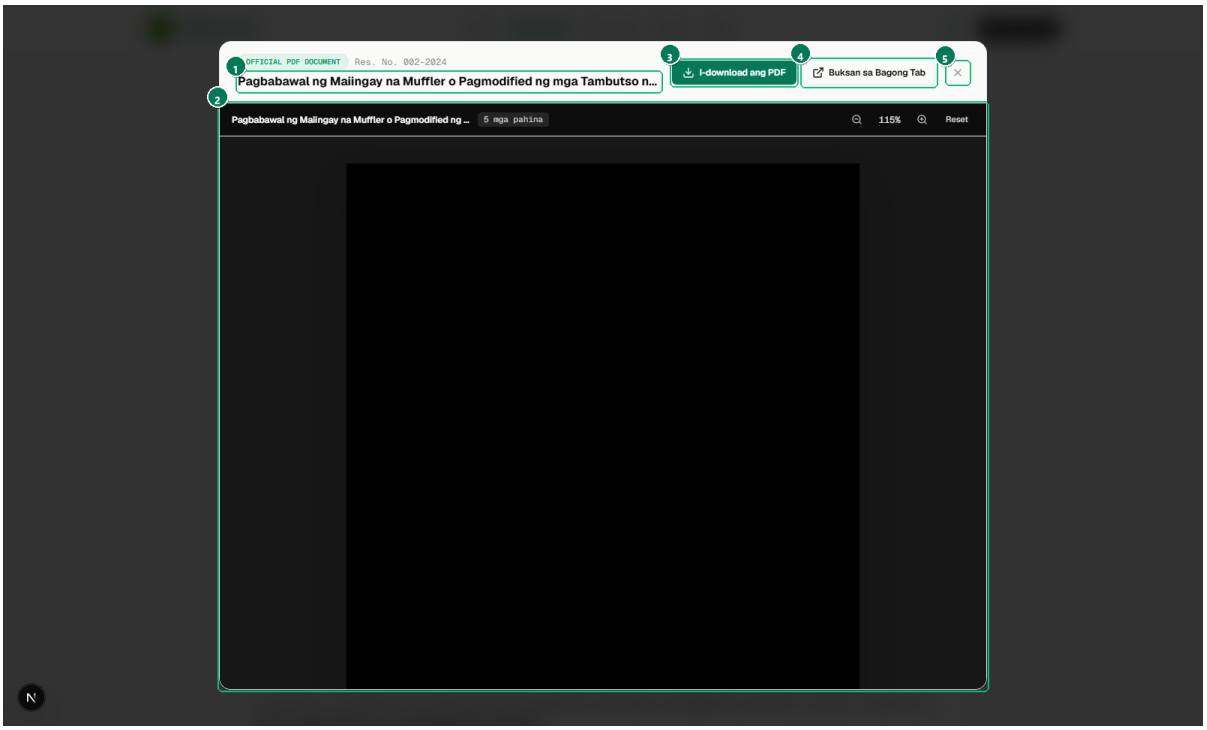


Figure 4.3: *Integrated in-browser document viewer allowing residents to examine official stamped ordinances, zoom into signatures, and download verified copies.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Official Document Modal Dialog Header & Identifier**	Official document modal dialog header displaying resolution title and unique code.
2	**Embedded Digital PDF Document Canvas**	Embedded document renderer presenting the original certified scanned paper copy with official seals.
3	**Direct PDF File Download Action Button**	Direct download button retrieving the official PDF file to the resident's local storage.
4	**Open in Dedicated Full-Screen Tab Button**	Open in dedicated browser tab action button for full-screen inspection and local printing.
5	**Modal Dismiss / Close Action (Esc)**	Accessible dismissal control triggering clean modal close (also supports keyboard Escape key).

4.4 AI Civic Assistant (Bilingual Gemini RAG)

Step 1: Navigating to AI Assistant — Click 'AI Assistant' in the header navigation or open /chatbot.

Step 2: Noting the Civic Safety Disclaimer — Read the informational notice at the top (Badge 1) clarifying that the assistant provides informational citations and not legal advice.

Step 3: Using Preset Action Inquiries — Click any of the quick prompt chips (Badge 2) to explore common questions regarding curfew hours, waste burning, or traffic rules.

Step 4: Typing Natural Language Inquiries — Type your question into the chat input field (Badge 5) in English, Tagalog, or Taglish, then click Send (Badge 6).

Step 5: Reviewing Citation-Grounded Responses — The Gemini RAG pipeline queries local municipal ordinances and streams an answer citing specific resolution numbers and fine schedules (Badge 4).

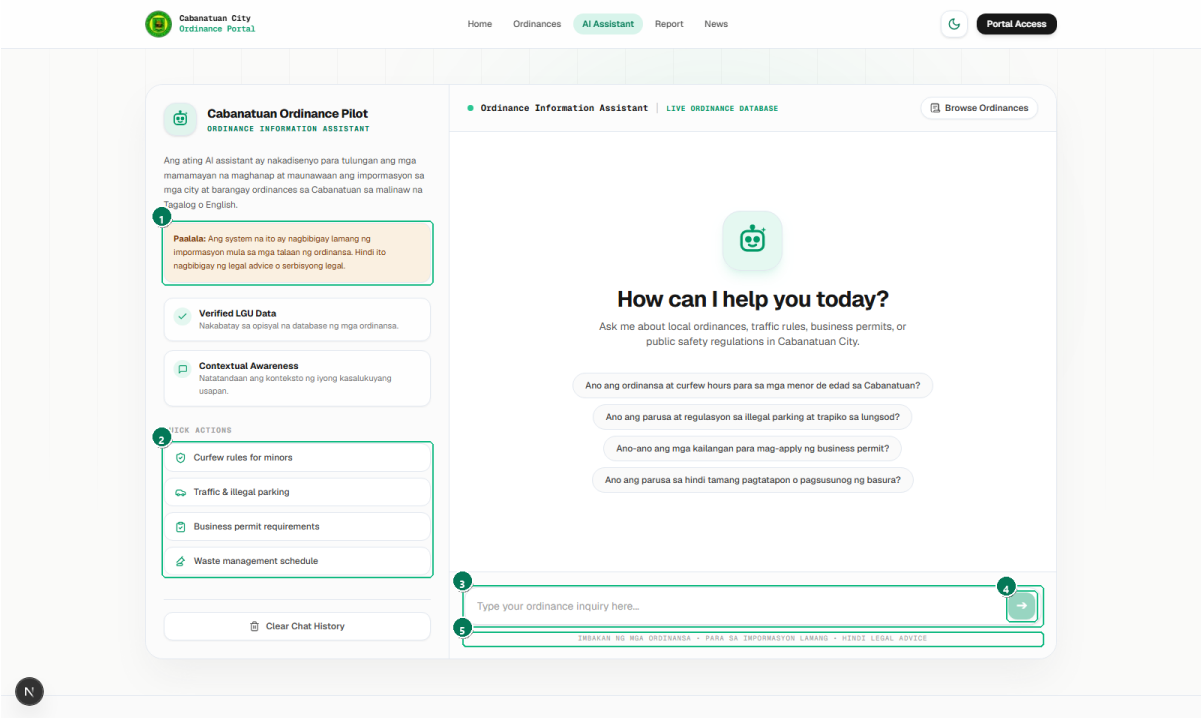


Figure 4.4: *Retrieval-Augmented Generation (RAG) assistant powered by Google Gemini, answering citizen inquiries with official ordinance citations.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Civic Safety & Informational Disclaimer Notice**	Prominent civic advisory banner stating AI outputs are informational and do not constitute formal legal counsel.
2	**Preset Quick Action Inquiry Buttons**	Preset citizen inquiry chips launching common questions regarding curfew, traffic, and waste management.
3	**Bilingual Question Input Field**	Bilingual prompt input supporting conversational inquiries with debounced token streaming.
4	**Send Message Action Button**	Message dispatch button transmitting prompt to the Gemini 3.5 Flash Lite retrieval pipeline.
5	**Official Informational Banner & Citation Notice**	Bottom civic disclaimer reminding citizens to verify official texts for formal legal proceedings.

4.5 Submitting a Community Incident Report

Step 1: Reviewing Emergency Hotline Notice — Inspect the prominent red emergency banner (Badge 1). If an incident involves an ongoing fire, medical emergency, or crime in progress, call the listed hotlines immediately (e.g., CDRRMO Rescue at 044-940-0161).

Step 2: Selecting Assigned Barangay — Open the Barangay selector (Badge 2) and select the specific barangay where the violation occurred (e.g., Camp Tinio).

Step 3: Choosing Violation Category — Select the category from the dropdown (Badge 3), such as 'Sanitation / Basura', 'Ingay / Loud Exhausts', or 'Obstruction'.

Step 4: Entering Subject & Details — Provide a brief summary in the Subject line (Badge 4) and full narrative details in the Textarea (Badge 5), including landmarks.

Step 5: Client-Side Draft Persistence — The form automatically saves typed content to browser local storage. If your page reloads accidentally, all typed information is instantly restored.

Step 6: Submitting the Report — Choose whether to remain anonymous (Badge 6) and click 'I-sumite ang Ulat sa Barangay' (Badge 7).

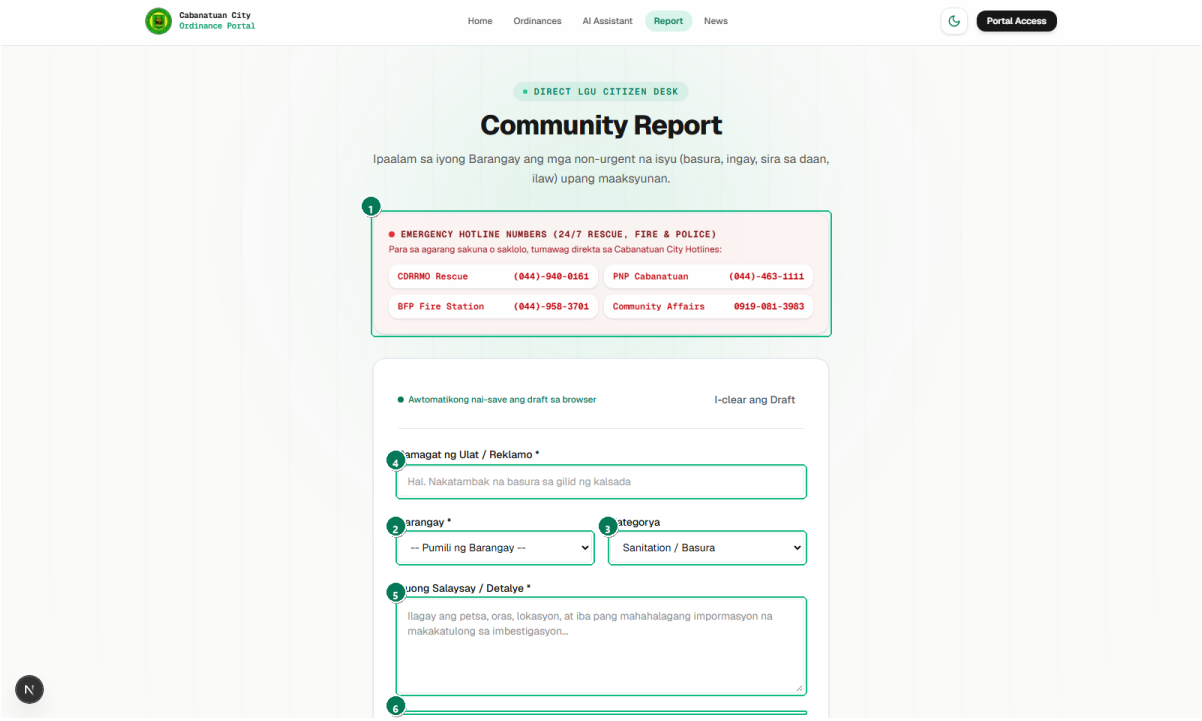


Figure 4.5: *Citizen incident submission interface equipped with client-side draft auto-save persistence and direct routing to barangay halls.*

#	Interface Element / Control	Operational Purpose & Usage
1	**24/7 City Emergency Hotline Numbers Banner**	Emergency numbers bulletin emphasizing that urgent rescue, fire, and police matters must call 911 / CDRRMO.
2	**Assigned Barangay Jurisdiction Dropdown**	Participating barangay dropdown routing the incident directly to the designated barangay hall.
3	**Incident Category Selector**	Standardized incident category selector (Open Burning, Noise Disturbance, Road Obstruction, etc.).
4	**Incident Subject / Summary Input Field**	Incident subject input capturing a concise summary of the observed ordinance violation.
5	**Incident Narrative & Location Details Textarea**	Narrative details textarea capturing factual descriptions of the observed ordinance violation.
6	**Anonymous Reporting Toggle**	Anonymous submission checkbox protecting resident identity with optional follow-up phone entry.
7	**Submit Incident Report Action**	Formal report submission trigger validating required fields and

	Button	committing to the database.

4.6 Submission Feedback & Tracking Reference

Step 1: Reviewing Confirmation Alert — Upon submission, a prominent green confirmation alert banner appears on screen (Badge 1).

Step 2: Verifying Destination Routing — The banner confirms that the incident has been officially routed to the target barangay hall (Badge 2).

Step 3: Recording Tracking Code — Copy or screenshot the generated unique Transaction Reference Code (Badge 3) for future follow-up with barangay officials.

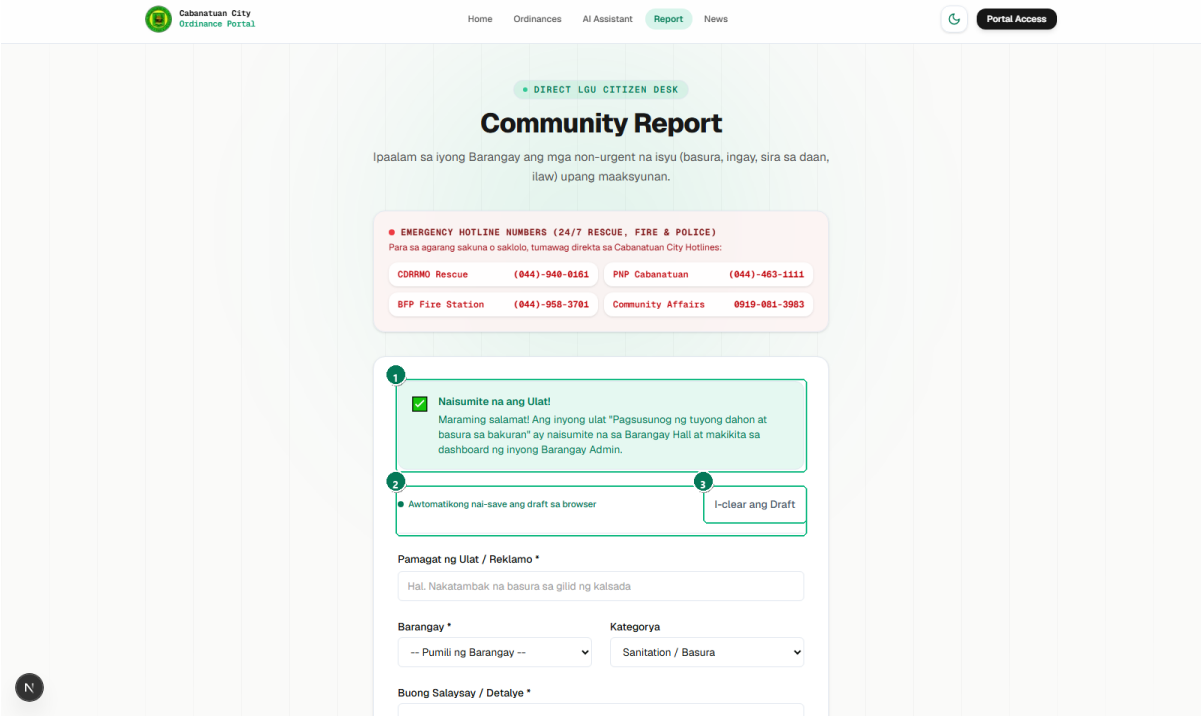


Figure 4.6: *Confirmation alert providing a unique tracking code for citizen verification, seamlessly connecting to barangay admin triage.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Official Submission Confirmation Alert Banner**	Official confirmation alert banner verifying direct submission to the target barangay hall.
2	**Draft Auto-Save Status Bar**	Client-side draft persistence status confirming the form state has been safely cleared.
3	**Reset & Clear Form Action Button**	Reset action button allowing the resident to clear form fields and prepare another submission.

4.7 Civic News & Public Advisories Feed

Step 1: Accessing Newsroom — Click 'Balita' in the navigation bar or visit /news.

Step 2: Filtering by Category — Toggle between 'Lahat', 'Lungsod', and 'Barangay' filters (Badge 2).

Step 3: Reading Pinned Urgent Directives — Inspect high-priority executive directives marked with the Pinned tag (Badge 3) at the top of the feed.

Step 4: Browsing Community Articles — Read chronological updates, cleanup drives, and municipal council session summaries (Badge 4).

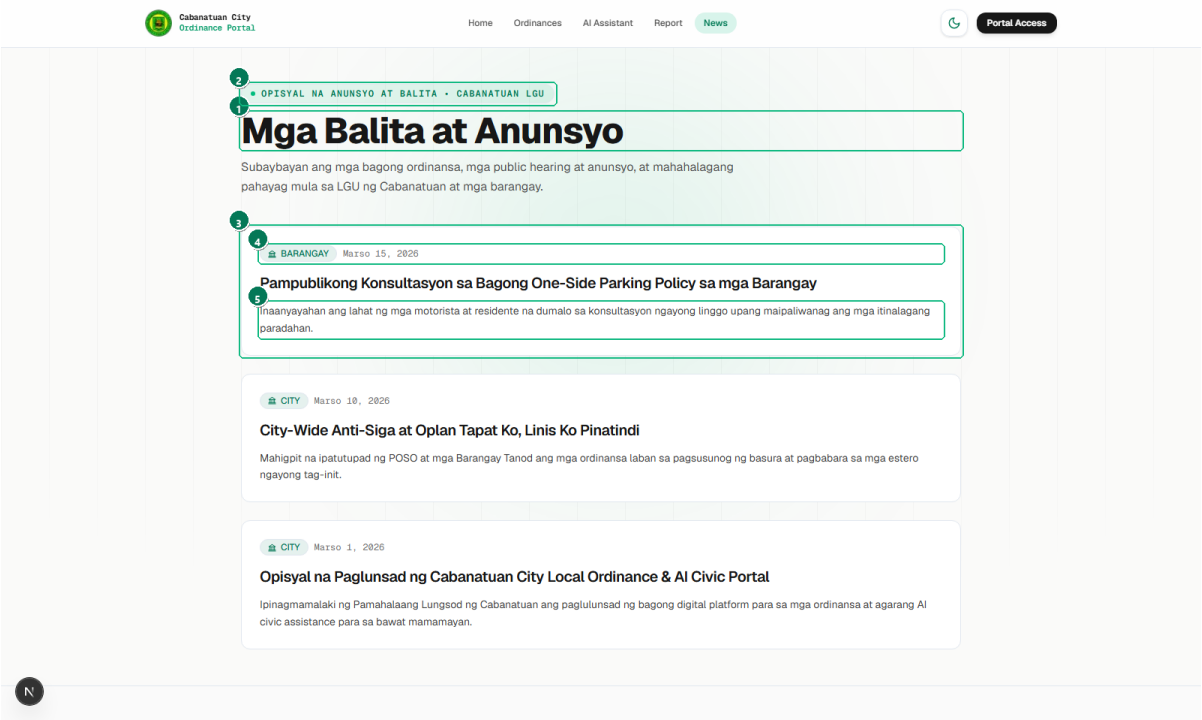


Figure 4.7: *Public information broadcast feed synchronizing official legislative reminders and community announcements across Cabanatuan City.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Civic News & Legislative Advisories Header**	Civic newsroom header displaying verified announcements from City Hall and participating barangays.
2	**Official LGU Public Authority Badge**	Official Cabanatuan LGU public authority badge validating authenticity of published notices.
3	**Official Executive Advisory Card**	Prominent official executive advisory card highlighting priority legislative announcements.
4	**Publication Date & Civic Category Badge**	Official timestamp and policy categorization badge denoting when the notice was enacted.
5	**Executive Directive Content Narrative**	Comprehensive text narrative detailing civic guidelines, meeting schedules, and public directives.

Chapter 5: Barangay Administrator Operations Guide

Barangay Administrators (Punong Barangays, Barangay Secretaries, and Sangguniang Barangay Members) possess dedicated administrative privileges scoped strictly to their registered barangay. Officials can draft local ordinances, parse scanned documents via Google Gemini multimodal OCR, transmit legislation to City Hall for legal review, triage community incident reports, and generate formal legislative reports.

5.1 Local Governance Dashboard Overview

Step 1: Signing In — Sign in using your verified barangay credentials (e.g., kapitan@cabanatuan.gov.ph / kapitan123).

Step 2: Verifying Jurisdiction Header — Confirm the header reads 'Barangay Governance Dashboard — Camp Tinio' (Badge 1). Data isolation prevents visibility of unrelated barangays.

Step 3: Inspecting Local KPI Cards — Monitor active local ordinances (Badge 2), pending proposals awaiting Sangguniang Panlungsod approval (Badge 3), and citizen reports (Badge 4).

Step 4: Triage Overview Feed — Examine incoming incident reports from constituents in the central activity table (Badge 5).

Step 5: Navigating Modules — Use the dedicated sidebar (Badge 6) to transition between Ordinances, Reports, and Analytics.

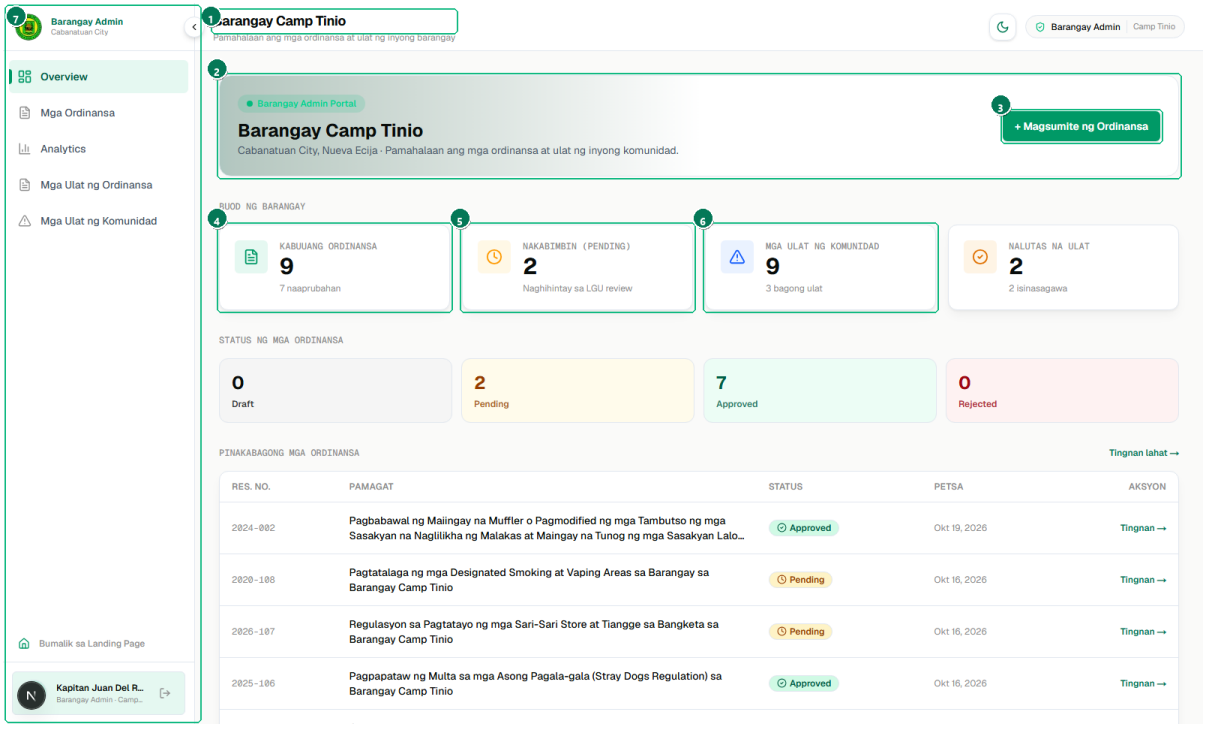


Figure 5.1: *Localized dashboard scoped exclusively to Barangay Camp Tinio, displaying local ordinance counts, pending approvals, and constituent reports.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Barangay Governance Dashboard Header**	Scoped header confirming administrative session for Barangay Camp Tinio with localized authority.

2	**Local Governance Welcome Banner**	Local governance welcome banner summarizing current operational status for the barangay council.
3	**Submit New Ordinance Quick Action Button**	Prominent quick-action button initiating a new barangay ordinance proposal transmittal.
4	**Local Enacted Ordinances Metric Card**	Active local ordinances counter tracking enacted barangay resolutions within the jurisdiction.
5	**Pending Legislative Proposals Metric Card**	Pending proposals counter monitoring legislation submitted to City Hall awaiting LGU approval.
6	**Active Community Incident Reports Metric Card**	Community reports KPI displaying incoming violation reports filed by barangay residents.
7	**Barangay Admin Dedicated Navigation Sidebar**	Barangay navigation sidebar providing rapid access to Ordinances, Community Reports, and Analytics.

5.2 Barangay Ordinance Management & Status Tabs

Step 1: Navigating to Repository — Click 'Mga Ordinansa' on the sidebar or visit /admin/barangay/ordinances.

Step 2: Triage by Status Tabs — Use the accessible status tabs (Badge 1) to inspect All, Pending, Approved, Draft, and Rejected ordinances.

Step 3: Searching Local Records — Type resolution numbers or keywords into the local search bar (Badge 3).

Step 4: Initiating New Enactment — Click the 'Magdagdag ng Ordinansa' button (Badge 2) to launch the proposal modal.

Step 5: Managing Legislative Records — Click action buttons (Badge 5) on any row to view, edit drafts, or resubmit rejected items.

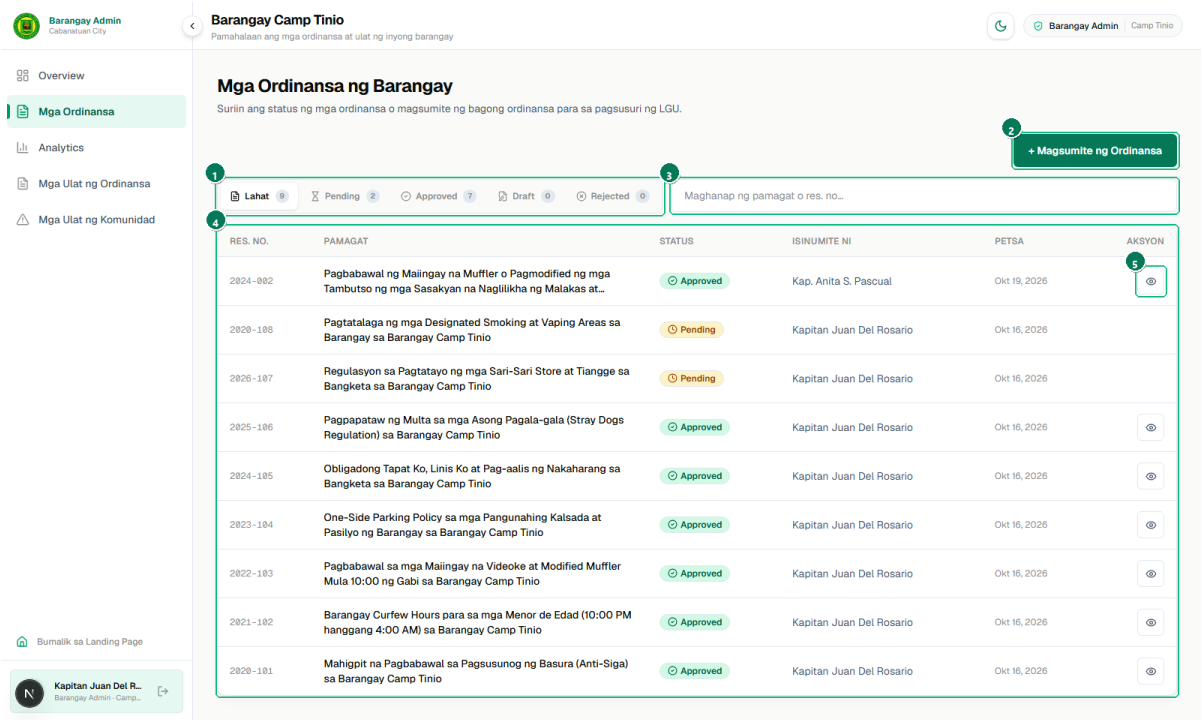


Figure 5.2: *Administrative table for managing local resolutions across their complete lifecycle: Drafts, Pending LGU Submissions, and Approved laws.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Legislative Status Tabs (Lahat, Pending, Approved, Draft, Rejected)**	Accessible status tabs filtering local records by All, Pending, Approved, Draft, and Rejected.
2	**Create / Propose New Ordinance Action Button**	Primary enactment button opening the multi-stage ordinance proposal and drafting modal dialog.
3	**Local Ordinance Keyword Search Input**	Search input enabling quick lookup of local barangay resolutions by keyword or series number.
4	**Barangay Ordinance Records Table with Status Badges**	Comprehensive barangay ordinance data table displaying resolution numbers, titles, and dates.
5	**Item Action Controls (View, Edit Draft, Resubmit)**	Action controls allowing officials to edit drafts, inspect rejected feedback, and resubmit items.

5.3 Drafting & Transmitting Local Ordinances

Step 1: Filling Legislative Metadata — Enter the proposed title and resolution number (Badge 2), followed by the policy category and year series (Badge 3).

Step 2: Drafting Legal Text & Sanctions — Enter legal provisions, prohibited acts, and fine schedules in the description and content inputs (Badge 4).

Step 3: Saving as Draft (Internal Staging) — Click 'I-save bilang Draft' (Badge 5). The record is stored with status DRAFT. City Hall cannot see or approve drafts until transmitted.

Step 4: Official Transmittal to LGU — Once approved by the Barangay Council, click 'Isumite sa LGU para sa Pagsusuri' (Badge 6). Status changes to PENDING and appears on the City Hall docket.

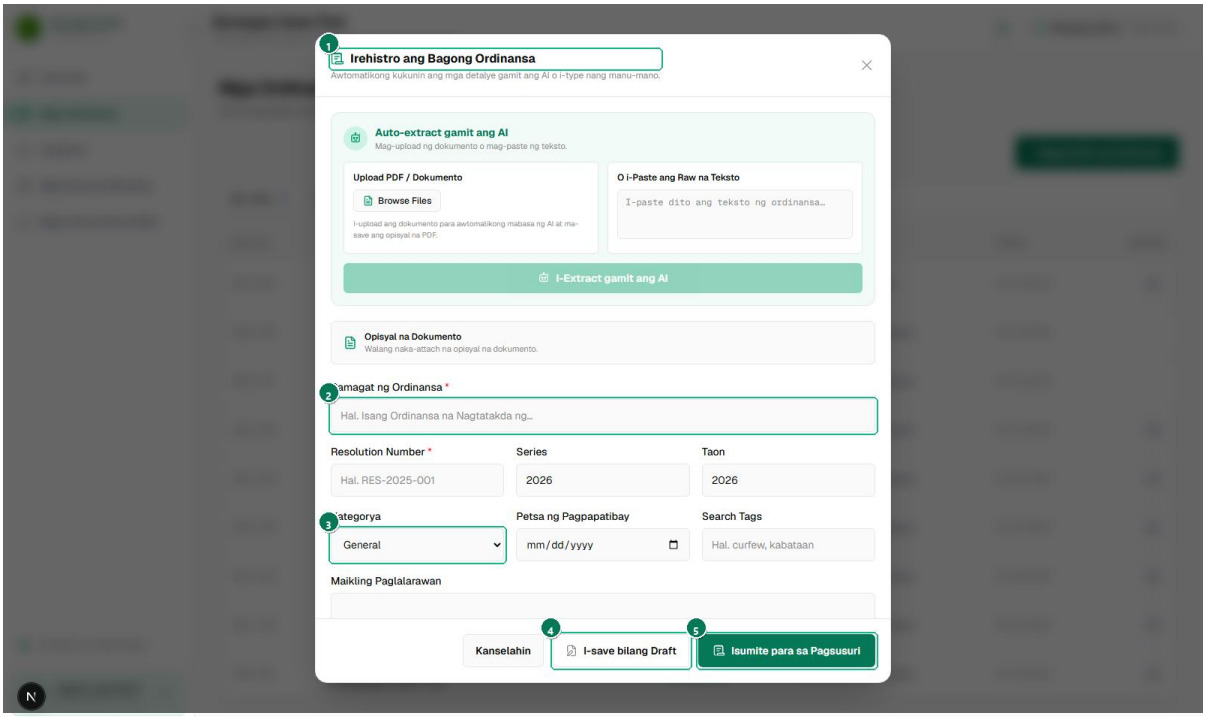


Figure 5.3: *Multi-stage proposal form enabling officials to prepare local legislation, save works-in-progress as drafts, or transmit them directly to City Hall.*

#	Interface Element / Control	Operational Purpose & Usage
1	**New Ordinance Proposal Modal Dialog Header**	Ordinance drafting modal header indicating proposal preparation for barangay council consideration.
2	**Proposed Ordinance Title Input Field**	Proposed ordinance title input adhering to standard barangay legislative naming conventions.
3	**Legislative Category Dropdown**	Legislative category selector mapping the ordinance to city governance domains.
4	**Save as Draft Action Button (Internal Staging)**	Draft saving action button persisting work-in-progress to the database without alerting City Hall.
5	**Submit to LGU for Approval Button (Official Transmittal)**	Official transmittal button submitting the finalized ordinance to the LGU Super Admin for legal review.

5.4 Multimodal AI Document Extraction (Barangay)

Step 1: Activating AI Extractor — In the ordinance creation modal, locate the 'AI Auto-Extraction' section (Badge 1).

Step 2: Uploading Scanned Document — Drag and drop a scanned PDF or smartphone photograph of the signed paper ordinance into the dropzone (Badge 2).

Step 3: Executing AI Extraction — Click 'I-Extract gamit ang AI' (Badge 3). The system dispatches the file to the Google Gemini multimodal model.

Step 4: Monitoring Multi-Stage Progress — Watch the animated progress indicator (Badge 4) as the model parses OCR text, resolution numbers, and penalty clauses.

Step 5: Human-in-the-Loop Review — Inspect the automatically populated fields (Badge 5). Make minor adjustments if necessary, then click Save.

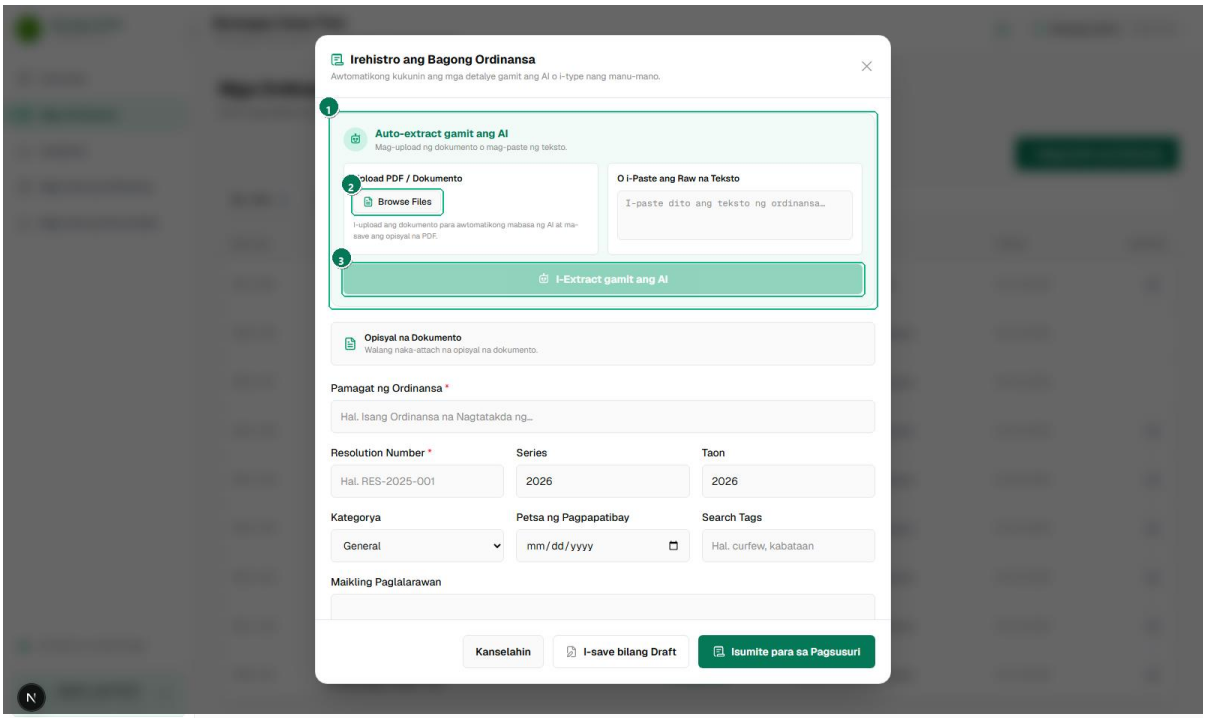


Figure 5.4: *Automated extraction tool utilizing Google Gemini to convert scanned physical paper ordinances into structured database records in seconds.*

#	Interface Element / Control	Operational Purpose & Usage
1	**AI Auto-Extraction Section Header & Gemini Sparkle Badge**	AI extraction module header featuring Google Gemini multimodal OCR integration badge.
2	**Scanned Document PDF / Image File Dropzone**	Drag-and-drop file upload zone accepting scanned PDF documents or multi-page high-resolution images.
3	**Execute Multimodal AI Parsing Button ('I-Extract gamit ang AI')**	Extraction execution trigger initiating automated text recognition and semantic clause extraction.

5.5 Constituent Incident Reports Desk

Step 1: Navigating to Reports Desk — Click 'Mga Ulat ng Residente' on the sidebar or visit /admin/barangay/reports.

Step 2: Filtering by Status — Click status filter buttons (Badge 2) to display New complaints, In Progress investigations, or Resolved incidents.

Step 3: Inspecting Incident Rows — Examine violation categories, landmark locations, and narrative testimonies submitted by residents (Badge 3).

Step 4: Verifying Anonymity Badges — Check whether the citizen filed anonymously or attached verified contact details (Badge 4).

Step 5: Launching Triage Dialog — Click the action button (Badge 5) on any row to open the case management modal.

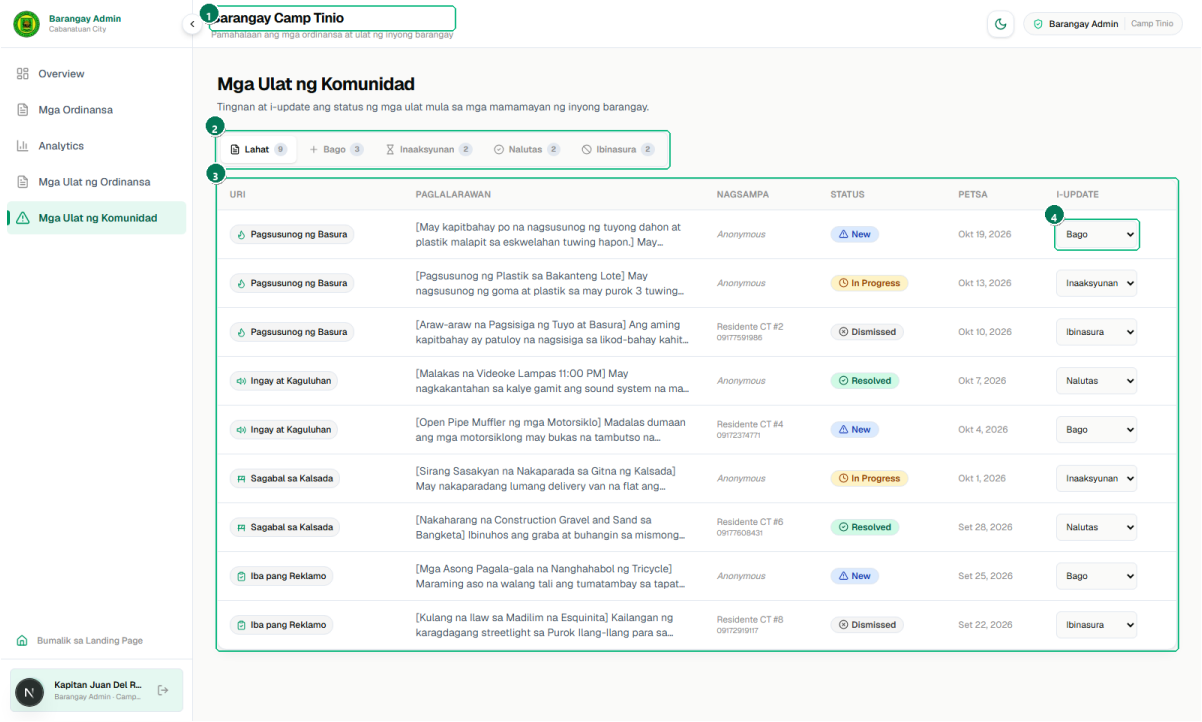


Figure 5.5: *Triage desk where barangay officials monitor, inspect, and process violation reports submitted by local residents.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Barangay Community Incident Reports Header**	Barangay reports desk header confirming oversight of citizen incident submissions in Camp Tinio.
2	**Status Filter Tabs (Lahat, Bago, Inaaksyunan, Nalutas, Ibinasura)**	Status filter buttons (All, New, In Progress, Resolved, Dismissed) segregating reports by stage.
3	**Incident Reports Records Table**	Incident reports records table displaying violations, landmarks, reporters, and submission dates.
4	**Inline Status Triage & Resolution Dropdown Control**	Inline status triage dropdown allowing immediate transition between New, In Progress, and Resolved.

5.6 Incident Case Triage & Status Resolution

Step 1: Reviewing Testimony & Evidence — Inspect the citizen's incident testimony, landmark coordinates, and timestamp (Badge 2).

Step 2: Updating Operational Status — Select the new status from the dropdown (Badge 3): change from 'Bago' (New) to 'Inaaksyunan' (In Progress) or 'Naresolba' (Resolved).

Step 3: Attaching Official Action Notes — In the Remarks field (Badge 4), enter official dispatch notes (e.g., 'Dispatched Barangay Tanod Team B for site inspection and mediation').

Step 4: Saving Updates — Click 'I-update ang Status' (Badge 5) to commit changes and log an audit timestamp.

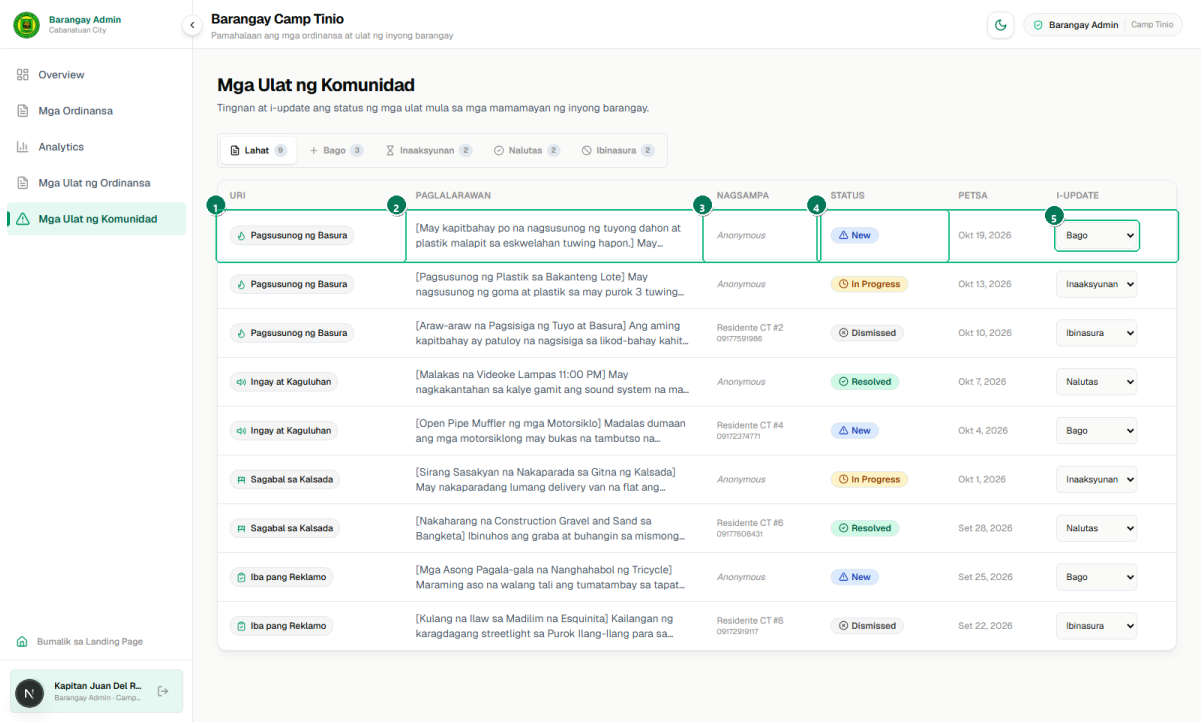


Figure 5.6: *Action dialog enabling barangay leaders to dispatch officers, record findings, and transition complaints to 'In Progress' or 'Resolved'.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Target Incident Report Record Row**	Focused incident report record row selected for investigation and administrative status update.
2	**Incident Classification Badge**	Violation category badge classifying the complaint (Trash Burning, Noise, Road Obstruction).
3	**Reporter Information & Anonymity Verification**	Reporter identification and contact details verification cell (with anonymous flag protection).
4	**Current Report Life-Cycle Status Badge**	Current life-cycle status badge reflecting real-time handling status in the central database.
5	**Official Status Mutation & Triage Dropdown Selector**	Administrative status mutation dropdown enabling officials to record action and resolve complaints.

5.7 Barangay Legislative Reporting & Export

Step 1: Navigating to Reporting Desk — Navigate to /admin/barangay/ordinance-reports or click 'Analytics & Reports'.

Step 2: Setting Report Parameters — Select report criteria, year ranges, and policy categories (Badge 2).

Step 3: Generating Summary Layout — Click 'I-generate ang Ulat' (Badge 3) to render the certified tabular layout.

Step 4: Exporting Formal PDF Documentation — Click 'I-download bilang PDF' (Badge 4) to compile a certified legislative record for the Barangay Hall physical archives.

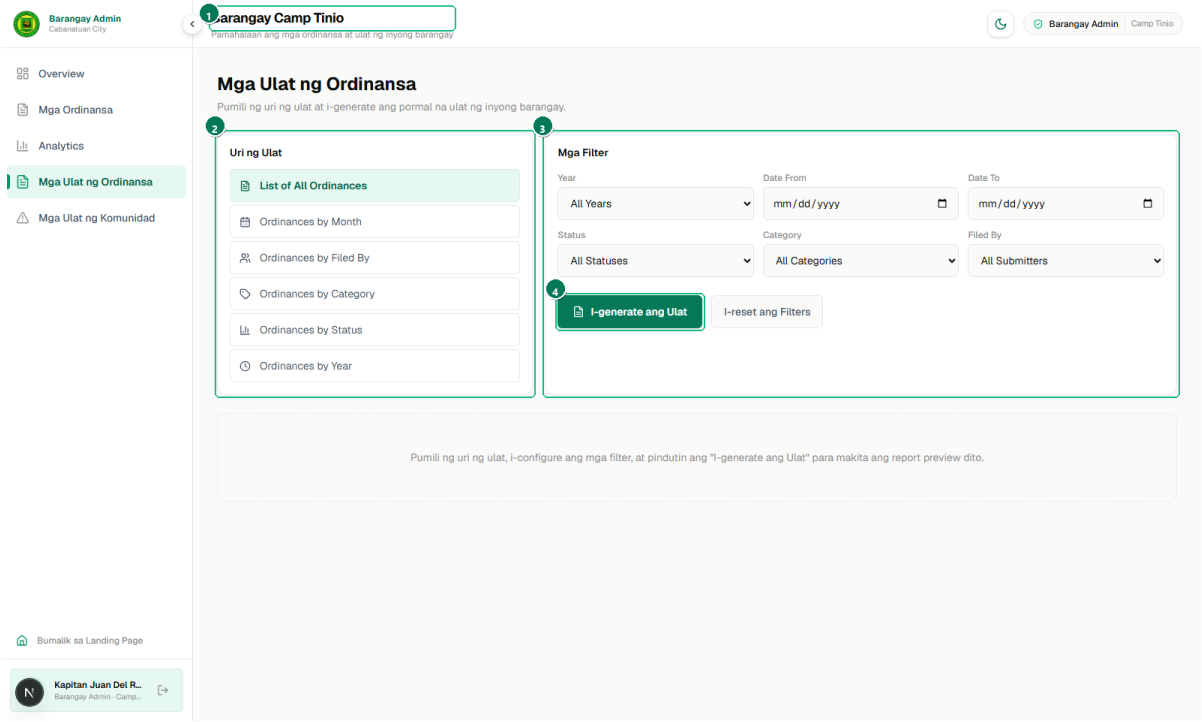


Figure 5.7: *Reporting module allowing barangay officials to generate formatted summaries of local legislation for barangay hall records.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Barangay-Scoped Legislative Reports Header**	Barangay legislative report generator header displaying localized documentation tools.
2	**Report Type Structure Selector Card**	Report type and filter selector defining parameters for generating formal barangay council reports.
3	**Multi-Criteria Filter Parameters Panel**	Multi-criteria filter panel with dropdowns for year series, status, category, and barangay scoping.
4	**Generate Localized Report Action Button**	Official export toolbar enabling instant download of formal PDF documentation with barangay headers.

Chapter 6: LGU Super Administrator Executive Guide

LGU Super Administrators (City Mayor, City Legal Council, Sangguniang Panlungsod Secretariat, and System Administrators) exercise full city-wide governance over all 89 barangays. Capabilities include managing the central municipal repository, reviewing and legally approving/rejecting barangay submissions, high-throughput AI document parsing, official account provisioning, city-wide incident auditing, civic broadcast publishing, and certified legislative PDF report generation.

6.1 Executive Governance Dashboard & City KPIs

Step 1: Executive Sign-In — Sign in with City Hall credentials (admin@cabanatuan.gov.ph / admin123) and navigate to /admin/lgu.

Step 2: Inspecting City-Wide KPI Cards — Examine live indicators (Badge 2): Total Ordinances (56), Approved Laws (43), Pending Barangay Proposals (6), Drafts (6), and Registered Barangays (14).

Step 3: Using Governance Quick Actions — Use direct action shortcuts (Badge 3) to enact city ordinances, generate certified reports, or jump directly to pending submissions.

Step 4: Auditing Real-Time Legislative Activity — Review the activity stream (Badge 4) for recent transmittals, approvals, and citizen report updates.

Step 5: Accessing Navigation Modules — The left sidebar (Badge 5) grants direct access to all eight administrative desks.

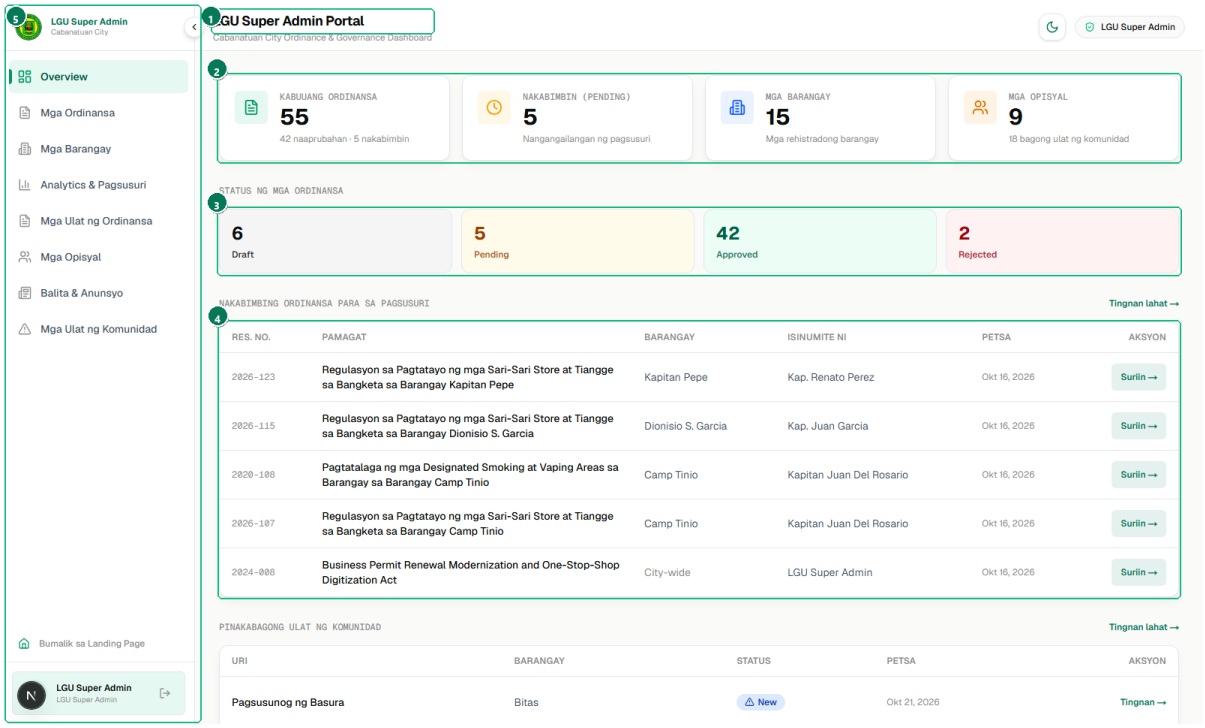


Figure 6.1: *Executive overview displaying city-wide legislative health metrics, pending barangay proposals, and real-time activity stream.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Executive Dashboard Header &**	City Government Executive Governance Dashboard header

	City Hall Governance Scope	confirming City Hall Super Admin privileges.
2	**City-Wide KPI Cards (Total Ordinances, Approved, Pending, Drafts, Barangays)**	Consolidated city-wide KPI cards showing Total Ordinances, Approved, Pending, Drafts, and Barangays.
3	**Ordinance Status Life-Cycle Summary Grid**	Executive quick-action shortcuts for rapid ordinance enactment, report generation, and pending reviews.
4	**Recent City-Wide Legislative Activity Log**	Recent legislative activity stream recording real-time enactment, submissions, and status mutations.
5	**Full Executive Navigation Sidebar (8 Integrated Administrative Modules)**	Full administrative sidebar providing direct routing across all eight executive management modules.

6.2 Central Ordinance Repository & Status Triage

Step 1: Navigating to Repository — Click 'Mga Ordinansa' on the sidebar or visit /admin/lgu/ordinances.

Step 2: Filtering by Status Tabs — Click accessible tabs with live record counters (Badge 1): All (56), Pending (6), Approved (43), Draft (6), Rejected.

Step 3: Performing Instant Search — Type search terms into the input box (Badge 2) for immediate client-side table filtering.

Step 4: Enacting City Legislation — Click 'Magdagdag ng Ordinansa' (Badge 3) to enact a new city-wide regulation or launch AI document parsing.

Step 5: Verifying Jurisdiction Badges — Inspect the origin badges (Badge 4) distinguishing municipal City ordinances from local Barangay submissions.

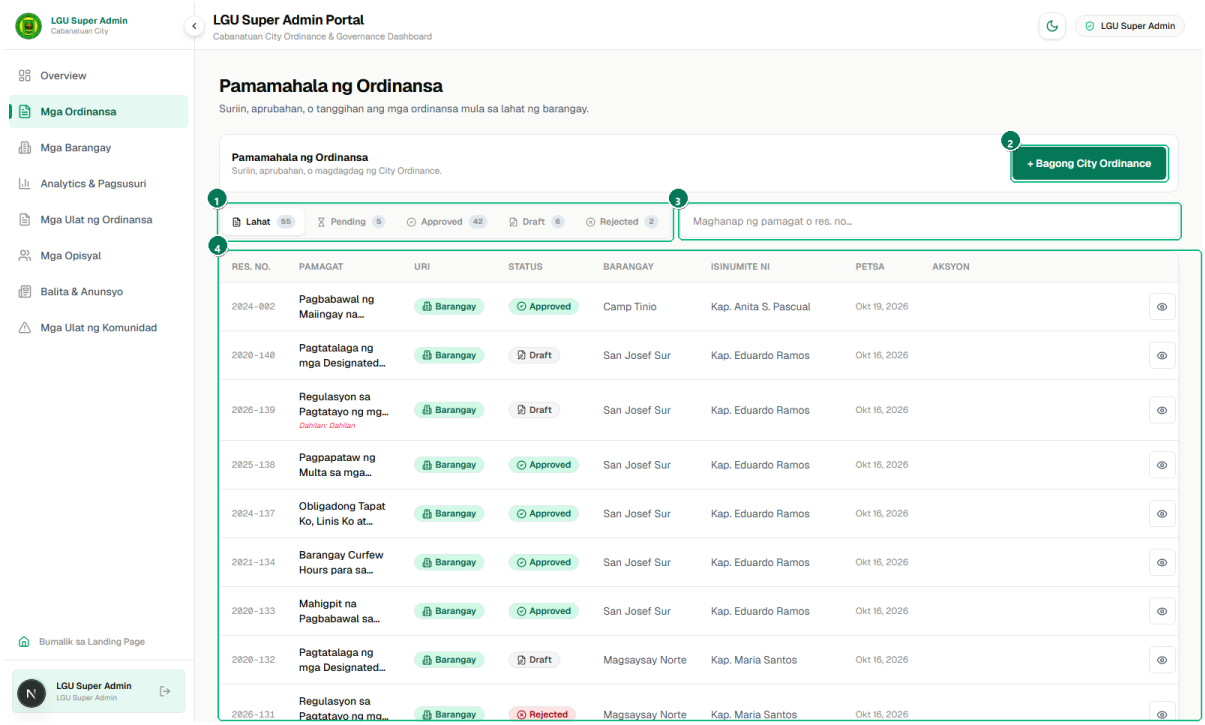


Figure 6.2: *Central repository managing all official city ordinances and legislative proposals submitted by eighty-nine administrative barangays.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Accessible Status Filter Tabs with Live Record Counters**	Status tabs with live record badges categorizing the complete municipal archive (56 total records).
2	**Add City Ordinance Action Button ('+ Bagong City Ordinance')**	New Ordinance enactment button launching the City Hall legislative creation and AI extraction modal.
3	**Instant City Ordinance Repository Search Input**	Client-side instant search input filtering city and barangay ordinances without network latency.
4	**City-Wide Ordinance Repository Table with Jurisdiction Badges**	Consolidated repository table with jurisdiction badges distinguishing City vs Barangay resolutions.

6.3 Reviewing & Processing Barangay Submissions

Step 1: Selecting Pending Submissions Tab — Click the 'Pending' tab on /admin/lgu/ordinances.

Step 2: Launching Review Modal — Locate a pending proposal (e.g., from Barangay Camp Tinio) and click 'Suriin / Review'.

Step 3: Evaluating Legal Provisions — Read the submitted title, background rationale, and proposed fine schedules (Badge 2).

Step 4: Attaching Legal Council Remarks — In the Remarks field (Badge 3), enter formal feedback (e.g., 'Reviewed and approved pursuant to City Council Session No. 14').

Step 5: Granting Enactment Approval — Click 'Aprubahan' (Badge 4). Status transitions immediately to APPROVED and the ordinance is published to the public portal.

Step 6: Requesting Revisions / Denying — Alternatively, enter required corrective actions and click 'Tanggihan' (Badge 5) to return the item to the barangay with attached notes.

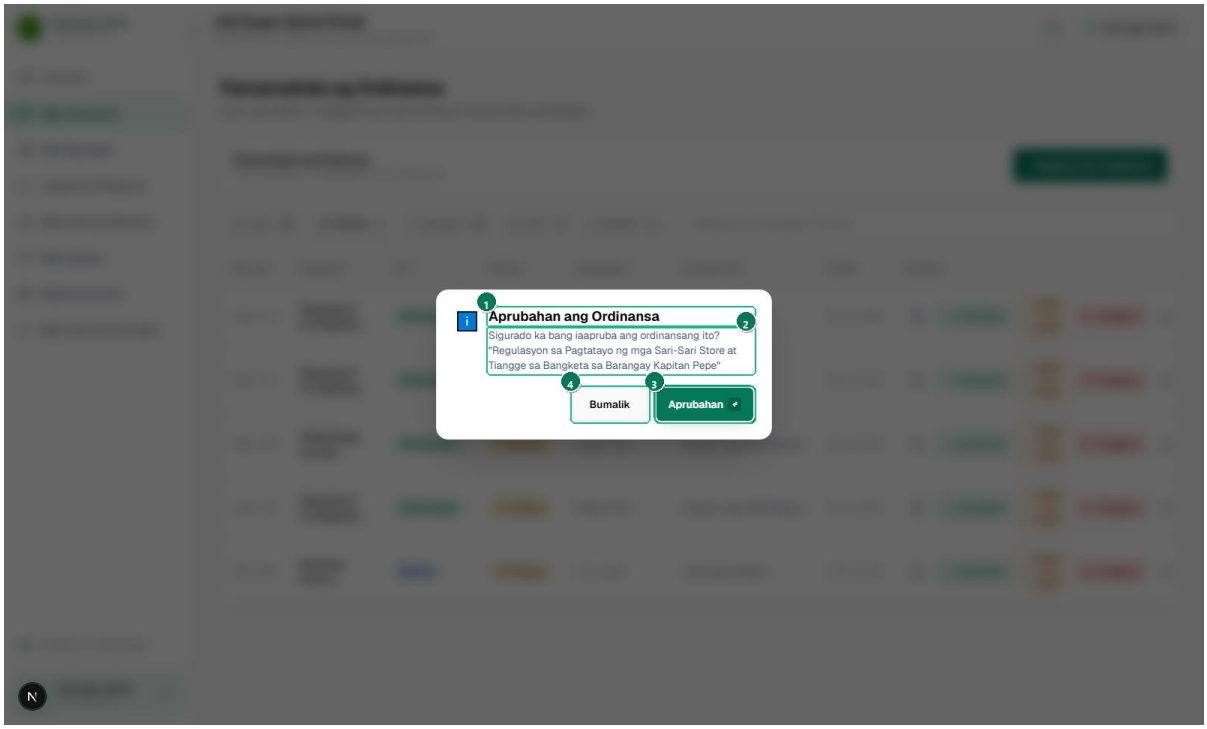


Figure 6.3: *Sangguniang Panlungsod review interface for evaluating proposed barangay ordinances, attaching official comments, and granting legal approval.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Sangguniang Panlungsod Legislative Approval Dialog Header**	Sangguniang Panlungsod Legislative Review modal displaying pending barangay proposal details.
2	**Pending Ordinance Title & Legal Enactment Notice**	Legal enactment confirmation notice asking the administrator to verify statutory compliance.
3	**Official Enactment / Approval Action Button ('Aprubahan')**	Official enactment approval button updating status to APPROVED and publishing to the public portal.
4	**Dismiss Review / Return Button ('Bumalik')**	Cancel / dismiss action button returning to the pending list without altering ordinance status.

6.4 Multimodal OCR Ordinance Extractor (Powered by Gemini)

Step 1: Opening City Ordinance Modal — Click 'Magdagdag ng Ordinansa' on /admin/lgu/ordinances.

Step 2: Uploading Official Scanned PDF — Drag and drop the scanned resolution file into the upload zone (Badge 2).

Step 3: Executing AI Multimodal Parsing — Click 'I-Extract gamit ang AI' (Badge 3). Google Gemini analyzes text and visual document structure.

Step 4: Inspecting Extracted Metadata — Verify auto-populated fields (Badge 4): Resolution Number, Series Year, Enacted Date, and Title.

Step 5: Validating Sanctions Breakdown — Verify structured first, second, and third offense fines (Badge 5), make minor text adjustments, and save.

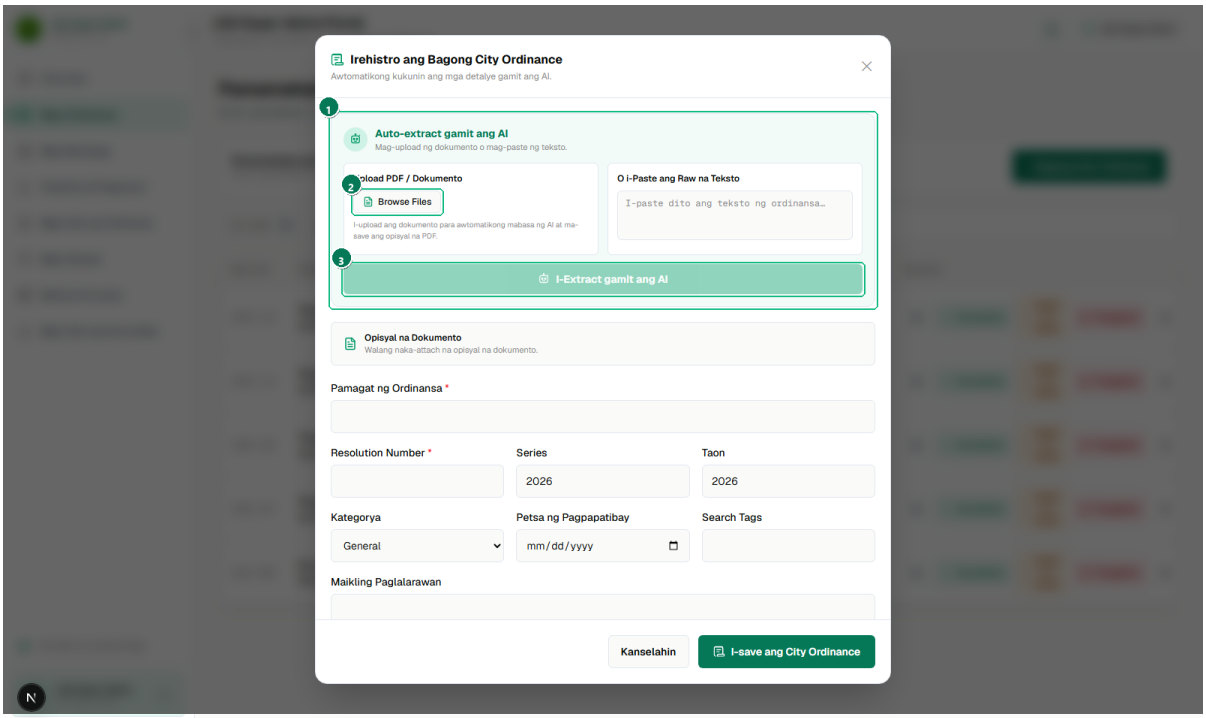


Figure 6.4: *High-throughput AI parser transforming scanned multi-page paper ordinances into verified digital records with tiered fine extraction.*

#	Interface Element / Control	Operational Purpose & Usage
1	**AI-Powered Multimodal OCR Ordinance Extractor Header**	LGU multimodal AI extraction tool powered by Google Gemini for scanning physical paper ordinances.
2	**Official Scanned Resolution Upload Dropzone**	Certified resolution document upload dropzone supporting multi-page PDF documents and high-res scans.
3	**Multimodal AI Parsing Execution Button**	AI extraction execution trigger initiating automated OCR, preamble stripping, and penalty categorization.

6.5 Administrative Accounts & Official Directory

Step 1: Navigating to Official Directory — Click 'Mga Opisyal / Users' on the sidebar or visit /admin/lgu/users.

Step 2: Registering a New Official — Click 'Magrehistro ng Opisyal' (Badge 2) to open the account provisioning dialog.

Step 3: Assigning Roles & Barangays — Select the role (LGU_ADMIN vs BARANGAY_ADMIN) and bind the account to their specific barangay jurisdiction (Badge 5).

Step 4: Security & Password Protection — Passwords are encrypted using salted bcrypt hashing before storage (Badge 3).

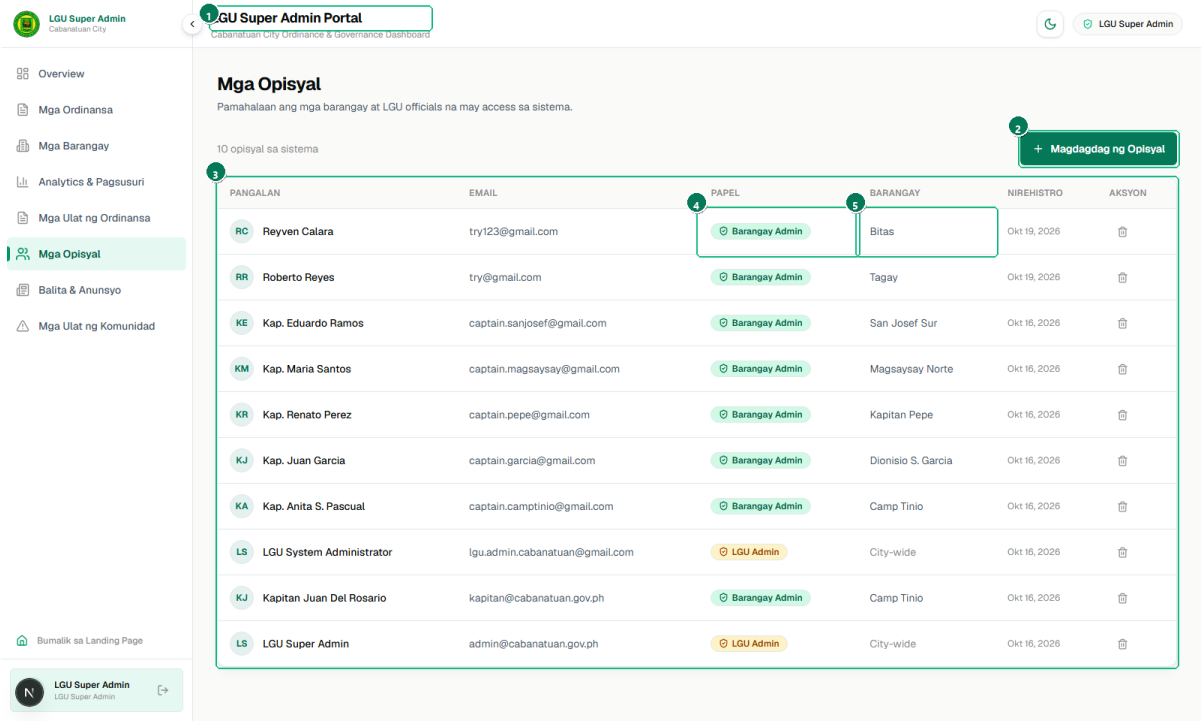


Figure 6.5: *Administrative user directory for provisioning barangay captains and city personnel with cryptographically secure role-based access control.*

#	Interface Element / Control	Operational Purpose & Usage
1	**City Government Official Personnel Roster Header**	Official personnel directory header displaying registered administrative accounts across Cabanatuan.
2	**Register New Official Action Button ('Magdagdag ng Opisyal')**	Provisioning action button opening the modal for registering new barangay captains and secretaries.
3	**Official Directory Table with Security Badges**	User directory data table showing full names, official emails, assigned roles, and creation dates.
4	**Role-Based Privilege Badges (LGU Admin vs Barangay Admin)**	Role privilege badges distinguishing LGU Super Admin permissions from scoped Barangay Admin rights.
5	**Assigned Territorial Barangay Jurisdiction Tag**	Territorial assignment badge ensuring barangay officials are strictly mapped to their local jurisdiction.

6.6 Barangay Directory & Territorial Coverage

Step 1: Navigating to Barangay Directory — Click 'Direktoryo ng Barangay' or visit /admin/lgu/barangays.

Step 2: Inspecting Participation Indicators — Examine active participation badges across the 14 registered barangays (Badge 2).

Step 3: Monitoring Enacted Legislation Counts — Inspect individual barangay cards (Badge 3) showing local resolution counts and active captains.

Step 4: Direct Barangay Archive Navigation — Click the view link on any card (Badge 4) to inspect that barangay's local archive.

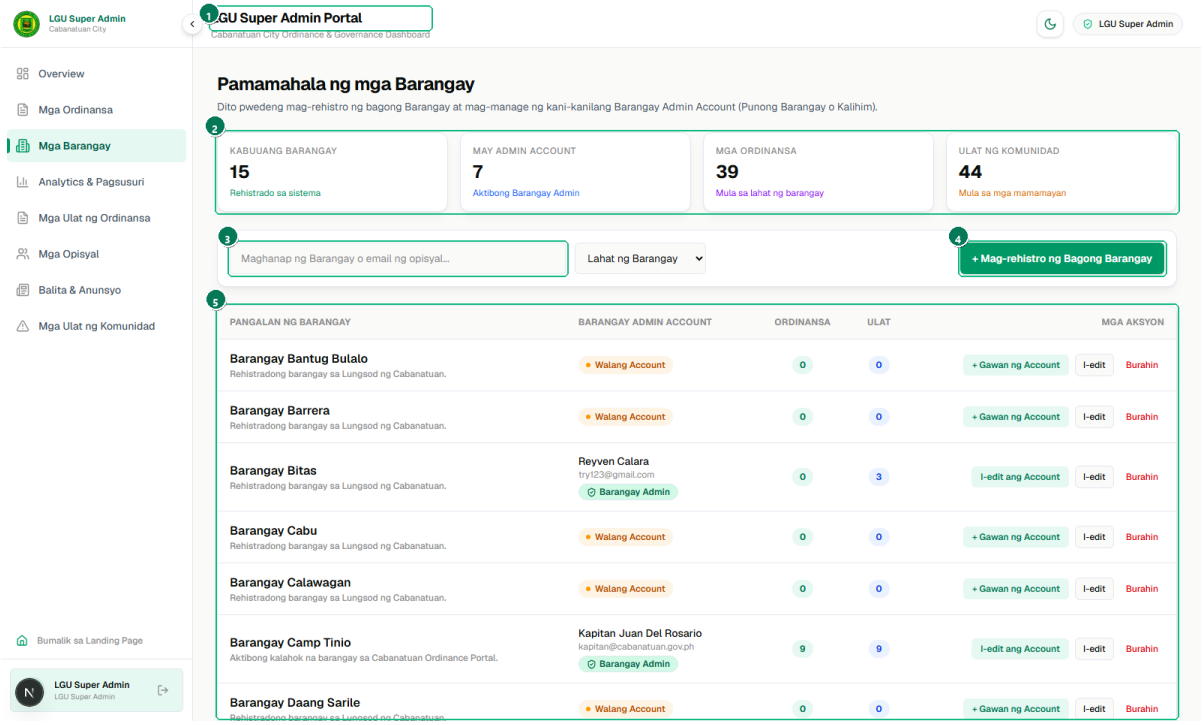


Figure 6.6: *Directory tracking participation, enacted local resolutions, and administrative contacts across all eighty-nine barangays of Cabanatuan City.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Cabanatuan City Barangay Directory Header**	Cabanatuan City Barangay Directory header tracking administrative coverage across all 89 barangays.
2	**Barangay Participation & Onboarding Indicators**	Participation and coverage statistics banner displaying total barangays and active admin accounts.
3	**Barangay Search & Filter Input Field**	Barangay search input field allowing instant lookup by barangay name or captain email address.
4	**Register New Barangay Action Button**	Primary registration button launching the new barangay creation and official account setup modal.
5	**Barangay Directory Table with Ordinance & Report Counts**	Consolidated directory records table displaying enacted local ordinances and resident report tallies.

6.7 City-Wide Community Incident Reports Desk

Step 1: Navigating to Reports Desk — Click 'Mga Ulat ng Komunidad' on the sidebar or visit /admin/lgu/reports.

Step 2: Filtering by Barangay Jurisdiction — Use the Barangay Dropdown (Badge 2) to filter complaints across any of Cabanatuan's barangays.

Step 3: Auditing Resolution Performance — Toggle status filter tabs (Badge 3) to identify unaddressed citizen reports.

Step 4: Inspecting Consolidated Evidence — Review violation categories, descriptions, and photographic evidence (Badge 4).

Step 5: Executive Override & Intervention — Click the status action control (Badge 5) to update status or assign city-level enforcement.

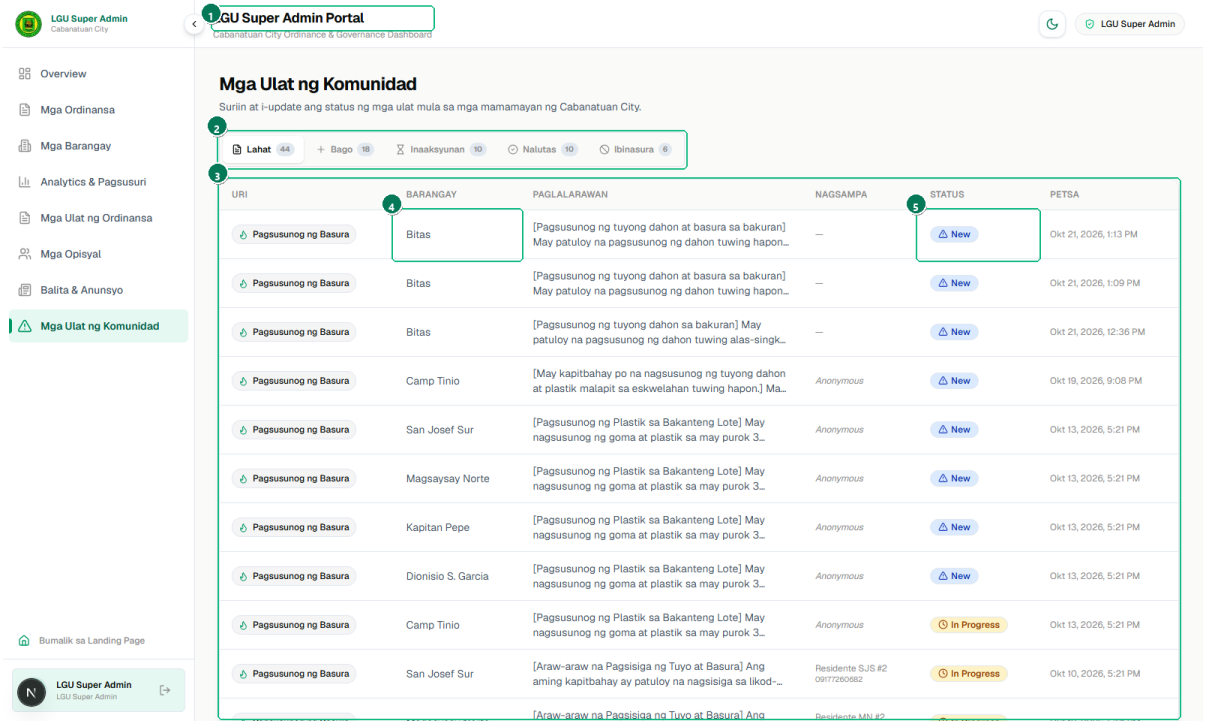


Figure 6.7: *Centralized supervisory desk enabling city hall administrators to monitor resident concerns and audit barangay action resolution times.*

#	Interface Element / Control	Operational Purpose & Usage
1	**City-Wide Community Reports Monitoring Header**	City-wide community incident monitoring desk header providing executive oversight of citizen reports.
2	**Resolution Status Filter Tabs (Lahat, Bago, Inaaksyunan, Nalutas, Ibinasura)**	Consolidated status tabs segregating reports across New, In Progress, Resolved, and Dismissed stages.
3	**Consolidated City-Wide Incident Records Table**	Comprehensive incident records table displaying violation categories, locations, and barangay assignments.

4	**Originating Barangay Jurisdiction Identification**	Originating barangay jurisdiction column indicating which local council has local primary authority.
5	**Current Operational Resolution Status Badge**	Real-time status badge tracking whether the barangay tanods or officials have addressed the issue.

6.8 Civic News & Public Advisories Broadcast Desk

Step 1: Navigating to News Desk — Click 'Mga Balita at Anunsyo' on the sidebar or visit /admin/lgu/news.

Step 2: Drafting New Public Advisory — Click 'Maglathala ng Anunsyo' (Badge 2) to open the announcement composer.

Step 3: Composing Title, Category & Content — Enter an informative title, select category (City or Barangay), and type announcement details.

Step 4: Pinning Urgent Public Advisories — Toggle the 'Pinned' switch (Badge 4) for critical weather, curfew, or public health directives.

Step 5: Publishing to Citizen Portal — Click Publish. The announcement synchronizes immediately with the public citizen portal (/news).

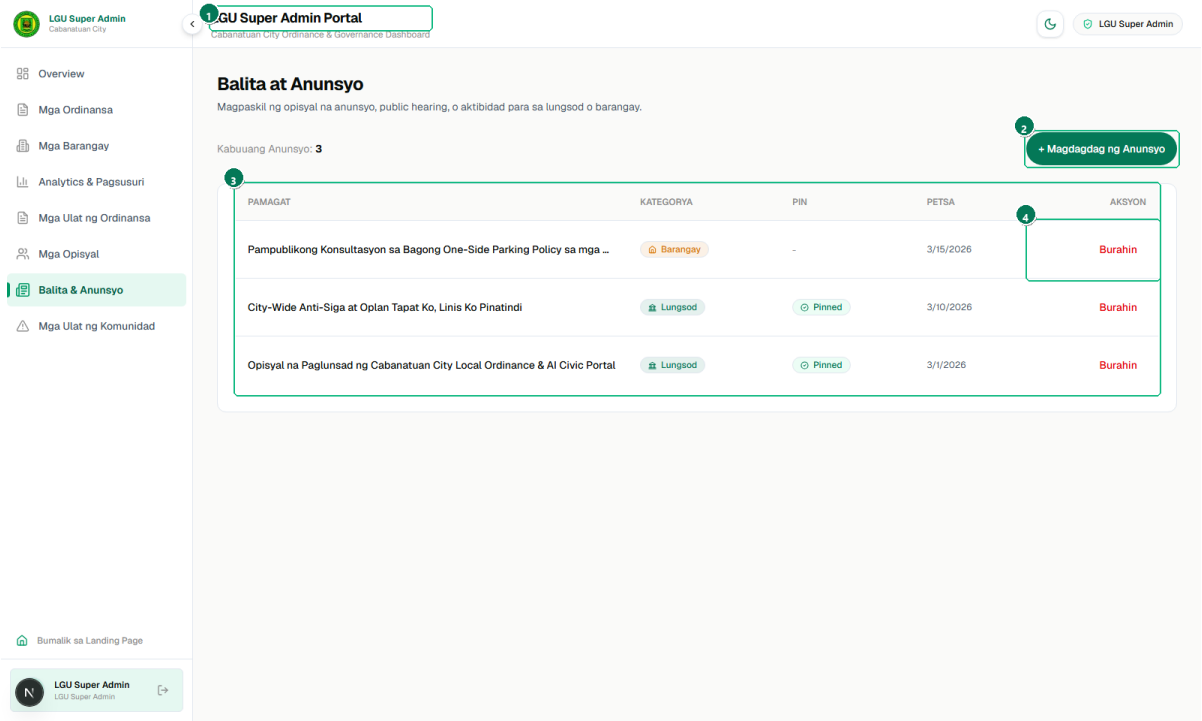


Figure 6.8: *Communication center for publishing legislative alerts, curfew reminders, and public health directives to the citizen-facing portal.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Civic News & Legislative Broadcasts Desk Header**	Civic news and public broadcast management desk header for City Hall legislative announcements.

2	**Publish New Advisory Action Button ('+ Magdagdag ng Anunsyo')**	Advisory publishing button opening the rich news composer dialog with category and pinning options.
3	**Published Public News & Advisories Table**	Published public advisories table displaying active executive notices, reminders, and press releases.
4	**Advisory Management Controls (Edit, Delete, Sync to Public Feed)**	Advisory management controls allowing city personnel to modify announcement content or archive items.

6.9 Legislative Report Generator & PDF Exporter

Step 1: Navigating to Report Generator — Click 'Mga Ulat at Dokumento' on the sidebar or visit /admin/lgu/ordinance-reports.

Step 2: Selecting Report Structure — Choose from eight specialized report structures (Badge 1): List of All Ordinances, by Month, Barangay, Filer, Category, Status, Type, or Year.

Step 3: Applying Criteria Filters — Set parameters in the filter panel (Badge 2) for Year, Status, Category, and Barangay origin.

Step 4: Executing Report Generation — Click 'I-generate ang Ulat' (Badge 3) to execute database aggregation and render the official preview.

Step 5: Inspecting Certified Document Preview — Verify the official Republic of the Philippines heading, Cabanatuan seal, and tabular format (Badge 4).

Step 6: Exporting Certified PDF Document — Click 'I-download bilang PDF' (Badge 5) to compile a certified legislative summary document.

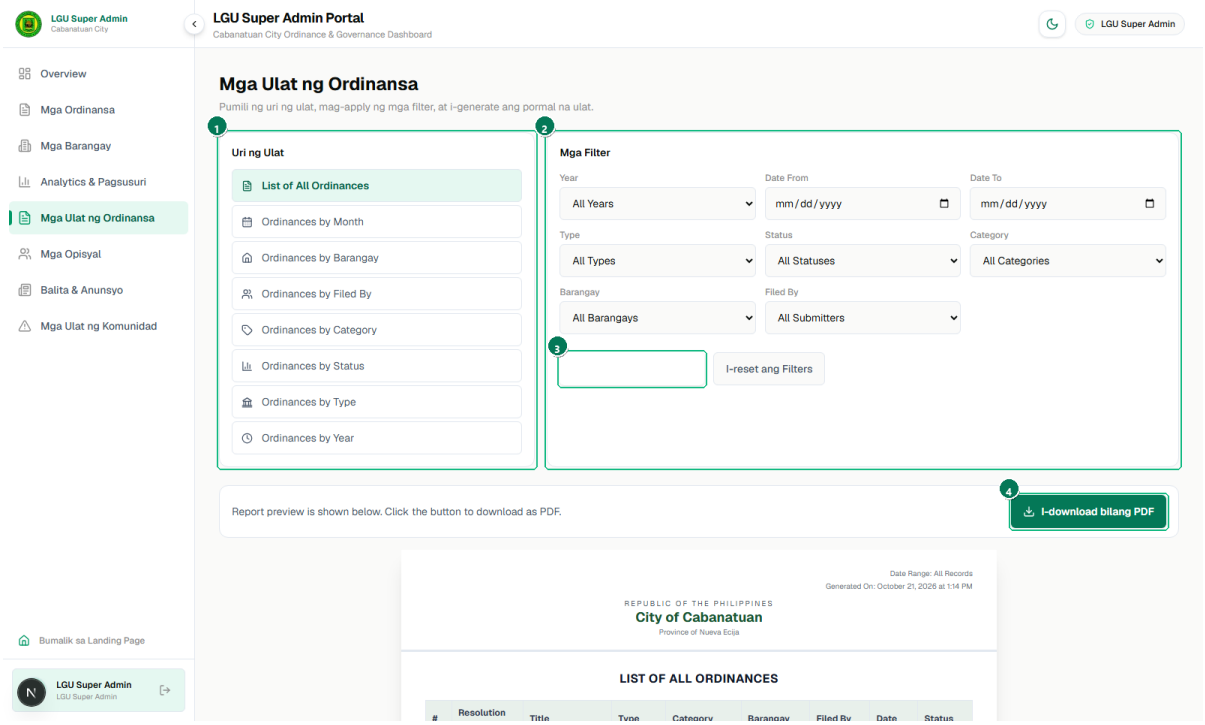


Figure 6.9: *Formal reporting tool generating certified legislative summaries with official headers, serial codes, and tabular PDF document generation.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Legislative Report Structure Selector Card (8 Standardized Structures)**	Report type selector offering 8 specialized structures (All, by Month, Barangay, Filer, Category, Status).
2	**Multi-Criteria Filter Parameters Panel (Year, Status, Category, Origin)**	Multi-criteria filter panel with dropdowns for year series, status, category, and barangay scoping.
3	**Execute Report Generation Action Button**	Report generation execution button querying the database and rendering the formal print-ready layout.
4	**Export Formal PDF Document Action Button ('I-download bilang PDF')**	Direct PDF download button generating a certified legislative summary document with pagination.

6.10 Governance Analytics & Policy Trend Visualizations

Step 1: Navigating to Analytics — Click 'Analytics' on the sidebar or visit /admin/lgu/analytics.

Step 2: Analyzing Enactment Velocity — Inspect the interactive line chart (Badge 2) tracking municipal ordinance enactment velocity over time.

Step 3: Examining Policy Distributions — Review the Category Distribution chart (Badge 3) showing thematic allocations (Health, Environment, Peace & Order).

Step 4: Auditing Barangay Participation — Inspect the Barangay Participation ranking (Badge 4) identifying top-contributing local councils.

Step 5: Tracking Community Resolution Efficiency — Analyze the Report Resolution Rate gauge (Badge 5) measuring average turnaround times for citizen complaints.

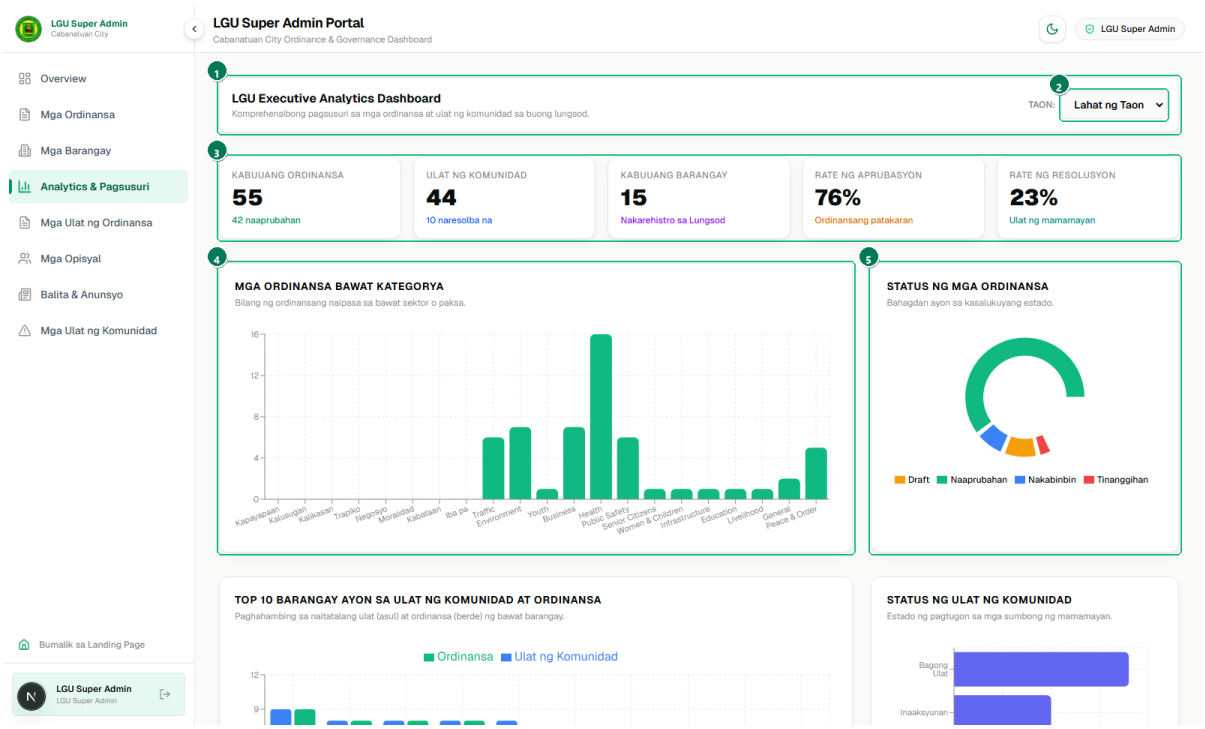


Figure 6.10: *Interactive data visualization suite tracking legislative velocity, thematic distributions, and civic report resolution performance.*

#	Interface Element / Control	Operational Purpose & Usage
1	**Governance Analytics & Policy Intelligence Header Card**	Governance analytics and policy intelligence header providing data-driven legislative insights.
2	**Legislative Year Filter Dropdown Selector**	Legislative year filter dropdown enabling dynamic scoping across calendar years or entire history.
3	**Executive Key Performance Indicators (KPI) Summary Banner**	Executive key performance indicator (KPI) metric banner showing total ordinances and resolution rate.
4	**Policy Category Distribution Breakdown Chart**	Policy category distribution chart illustrating legislative focus across health, safety, and environment.
5	**Legislative Status Breakdown Distribution Chart**	Legislative status proportion pie chart displaying approved, pending, draft, and rejected distribution.

Chapter 7: Data Isolation & Security Guidelines

To guarantee administrative integrity, operational privacy, and compliance with the Philippine Data Privacy Act of 2012 (RA 10173), the Cabanatuan City Ordinance Portal implements multi-tiered security protocols:

1. Multi-Tenant Jurisdictional Isolation:
Every database mutation (creation, status update, deletion) verifies that the authenticated user's session barangayId matches the target record's barangayId. Barangay Captains cannot view drafts or modify citizen reports belonging to neighboring barangays.

2. Cryptographic Token Protection:
Authentication tokens are signed with HMAC-SHA256 and transmitted exclusively in HTTP-only, SameSite=Lax cookies with path-level route guards in Next.js middleware, mitigating cross-site scripting (XSS) and CSRF vectors.

3. Salted Password Hashing:
User credentials are encrypted using salted bcrypt hashing with a work factor of 10. No plaintext credentials ever touch persistent storage or server log files.

4. Audit Logging & Legislative Traceability:
All ordinance approvals, rejections, and citizen report status updates record an immutable audit trail capturing the actor's user ID, timestamp, and review remarks for complete civic accountability.

> **PRO-TIP: IMMUTABLE LEGISLATIVE AUDIT TRAILS**
> When an LGU Super Administrator approves or denies a proposed ordinance, the system records: (1) reviewedById foreign key reference, (2) approvedAt timestamp, and (3) rejectedReason text notes. This audit record is preserved permanently in the database and visible in formal legislative PDF exports.

Chapter 8: Troubleshooting, Prerequisites & FAQ

8.1 System Requirements & Browser Compatibility

System Component	Technical Specification
Supported Browsers	Google Chrome 110+, Microsoft Edge 110+, Mozilla Firefox 115+, Safari 16+
Display Resolution	Responsive: 360px mobile viewport up to 1920x1080 Full HD desktop
Server Runtime	Node.js 20+ LTS or Node.js 22+, SQLite 3.40+, Better-SQLite3
Hardware Requirements	Minimum 4 GB RAM, 2 GHz dual-core processor, 500 MB disk space
Network Connectivity	Broadband internet connection required for Google Gemini API inference; offline database exploration supported locally

8.2 Common Operational Issues & Troubleshooting

⚠ **CAUTION: ISSUE 1: AI MULTIMODAL EXTRACTION TIMEOUT OR ERROR**
Symptom: Uploading a large scanned PDF results in an error message or spinning wheel.
Resolution: Ensure the uploaded PDF or image is under 15 MB and clearly legible. Verify that your GEMINI_API_KEY environment variable is configured in .env and that the server has an active internet connection to access the Google GenAI endpoints.

⚠ **CAUTION: ISSUE 2: FORBIDDEN / REDIRECT LOOP DURING LOGIN**
Symptom: Attempting to access /admin/lgu redirects back to /login.
Resolution: Clear your browser session cookies for localhost:3000. Re-authenticate using the exact LGU Super Admin credentials (admin@cabanatuan.gov.ph / admin123). Ensure that caps lock is off.

📌 **NOTE: ISSUE 3: SUBMITTED ORDINANCE NOT APPEARING ON PUBLIC EXPLORER**
Symptom: A barangay official submitted an ordinance, but citizens cannot find it in /ordinances.
Resolution: Newly submitted barangay ordinances carry the status 'PENDING'. They only appear publicly once an LGU Super Administrator reviews and officially APPROVES the proposal pursuant to Sangguniang Panlungsod procedures.

8.3 Defense Panel Technical FAQ

Q1: Why was Google Gemini chosen over OpenAI or self-hosted models?

Answer: Google Gemini provides superior multimodal vision for analyzing scanned physical documents with stamps and signatures, low inference latency, and excellent bilingual Tagalog/English comprehension. In our RAG architecture, all city data remains securely on our local database, invoking Gemini solely for context synthesis.

Q2: How does the system prevent AI hallucinations and unauthorized legal advice?

Answer: The AI Civic Assistant utilizes strict prompt grounding with local retrieval injection. It only answers questions based on retrieved municipal ordinances and refuses to answer out-of-scope national legal questions. A prominent civic disclaimer informs residents that outputs are informational only.

Q3: How is multi-tenant security handled across eighty-nine barangays?

Answer: We enforce multi-tenant role-based access control using Prisma ORM and JWT tokens. Each barangay session is strictly bound to its assigned barangayId, preventing unauthorized cross-jurisdiction access or modification.